

**INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS
ACADEMY (ICDFA)**

SBT-DF204 · Computer Forensics Case Studies

CASE STUDY 2

**Reconstructing Suspected Drug Trafficking From Chrome
Web History**

*Individual Forensic Report: SQL Evidence, Labelled Screenshots, UTC Timeline and
Evidence Log*

Field	Detail
Student	Ibrahim Disih Garba
Registration No.	2025/FWSD/11521
Programme	Fellowship in Web Application Security & Digital Forensics
Course	SBT-DF204: Computer Forensics Case Studies
Assessment	Case Study 2: Individual browser history investigation
Instructor	Aminu Idris, AMCPN
Batch	BATCH-B2025 · L1/S2
Evidence item	Chrome History SQLite database ("History"), SHA-256 b991b0fa69662c44...
Date of analysis / submission	9 October 2026

Acknowledgement

I thank my instructor, Aminu Idris, AMCPN, for the teaching materials on URL structure, redirection, user web activity and Chrome History analysis that framed this investigation. I also acknowledge the International Cybersecurity and Digital Forensics Academy (ICDFA) for providing the case dataset and the assessment brief.

Declaration

I, Ibrahim Diseh Garba (Registration No. 2025/FWSD/11521), declare that this report is my own work. The SQL analysis, interpretation, timeline and evidence log were produced on a hash-verified working copy of the supplied Chrome History database. I did not visit, open or interact with any live website, posting, email account, image link, exchange, wallet, address or transaction identifier referenced in the history. Session tokens and other data unrelated to the findings have been truncated or omitted from the figures; case-relevant identifiers have been retained so that the examiner can verify each finding.

Signature: _____ Date: 9 October 2026

Executive Summary

This report examines a Google Chrome History database supplied for SBT-DF204 Case Study 2. The case hypothesis is that the browser was used to advertise controlled drugs, communicate with an interested buyer, exchange payment information, verify a cryptocurrency transaction and retain proof of payment. The database was copied, hashed with SHA-256, and examined read-only using `sqlite3`. All 74 visit records, 57 URL records, one download record and its URL chain were reviewed. Chrome WebKit timestamps were converted to UTC, and each visit's `transition` bit-field was decoded into a core type and qualifier flags, then read together with `from_visit` and `opener_visit` to rebuild navigation chains.

All recorded activity falls within one window on **19 April 2022, 13:54:18 – 15:02:56 UTC** (68 minutes 38 seconds). Within that window the profile signed in to a Craigslist account and completed a seven-step form-submission workflow ending at a “manage posting” page for posting **7473121658**, then opened the public listing for the same posting ID, titled “cheaper than Rx supplements”, in the Baltimore health-and-beauty category. The profile then signed in to Gmail; page titles identify the account as **unsub.fscs@gmail.com** and show two threads, one titled “cheaper than Rx supplements - order” and one whose title contains Bitcoin transaction ID `517b2156...afdc5`. A link in the second thread led to an Imgur image, which Chrome saved as **proof_of_payment.png**. Within three minutes the same transaction ID was looked up on `blockchain.com` and, after a Kraken sign-in with device approval, opened from the Kraken funding page on `mempool.space`. The session ends with compose activity in the transaction thread.

These artefacts correlate across five independent sites through two shared identifiers (posting ID and transaction ID), which gives the reconstructed sequence considerable internal coherence. However, the History database records URLs, page titles, times and navigation context only. It does not preserve the advertisement text, email bodies, image content or exchange data, and it cannot identify the person at the keyboard. Confidence is therefore graded: **high** that this browser profile performed the recorded sequence; **moderate** that the sequence represents a sale-and-payment workflow; **low** that the goods were controlled drugs or that any named individual operated the profile.

Objectives

1. Acquire and preserve the supplied Chrome History database, recording its source, file name, size and SHA-256 value, and verify that analysis did not alter it.
2. Identify the relevant SQLite tables and fields, and explain how the `urls`, `visits`, `downloads` and `downloads_url_chains` tables relate.
3. Convert Chrome WebKit timestamps to UTC using a documented, reproducible method.
4. Interpret navigation using transition core types, qualifier bits, `from_visit` and `opener_visit` links.
5. Correlate posting, communication, payment-verification and download artefacts into a UTC timeline that separates observation from interpretation.
6. State a defensible conclusion with graded confidence, limitations and alternative explanations, including the distinction between browser-profile activity and human identity.

Table of Contents

Acknowledgement	2
Declaration.....	2
Executive Summary.....	3
Objectives	4
Table of Contents.....	5
1.0 Case Reference and Student Identification	7
2.0 Executive Conclusion	8
3.0 Evidence Acquisition and Integrity	9
3.1 Source of the Evidence	9
3.2 Working-Copy Method and Access Precautions	9
3.3 Chain-of-Custody Worksheet	10
3.4 Redaction Approach.....	10
4.0 Database Method	11
4.1 Tables Reviewed.....	11
4.2 WebKit Timestamp Conversion	11
4.3 Transition Interpretation	12
4.4 Query Set	13
5.0 Findings.....	14
Question 1: What evidence file was acquired and how was it preserved?	14
Question 2: Which browser-history tables and fields are relevant?	14
Question 3: When did the recorded web activity occur?	14
Question 4: How did the browser reach the material URLs?	14
Question 5: What evidence relates to the suspected sale or posting?	15
Question 6: What evidence relates to communications and payment activity?	16
Question 7: What is the reconstructed timeline?	19
Question 8: What conclusion can be defended?	19
6.0 Timeline (Summary, UTC).....	20
7.0 Attribution Assessment.....	21
7.1 Limitations and Alternative Explanations	21
8.0 Challenges Encountered.....	23
9.0 Recommendations.....	24
9.1 Further Investigative Steps	24
9.2 Examination Practice	24
10.0 Conclusion	25
11.0 References.....	26

12.0 Appendices.....	28
12.1 Evidence Log	28
12.2 Timeline Table (UTC)	29
12.3 Labelled SQL Output Screenshots.....	31
12.4 SQL Query Listing.....	43

Note: right-click the table and choose “Update Field” in Microsoft Word to refresh page numbers after editing.

1.0 Case Reference and Student Identification

Field	Detail
Full name	Ibrahim Diseh Garba
Registration number	2025/FWSD/11521
Course code	SBT-DF204: Computer Forensics Case Studies
Case title	Reconstructing Suspected Drug Trafficking From Chrome Web History
Case reference	SBT-DF204-CS2-2025-FWSD-11521
Evidence item	Google Chrome History SQLite database, original file name “History”
Time standard	UTC throughout; local time shown separately for reference only (Section 4.3)
Date of analysis	9 October 2026

2.0 Executive Conclusion

The Chrome profile represented by this History database was used on 19 April 2022 between 13:54:18 and 15:02:56 UTC to (i) create, confirm and view a Craigslist posting (ID 7473121658) titled “cheaper than Rx supplements”; (ii) view Gmail threads whose titles link that posting to an “order” and to Bitcoin transaction 517b2156...afdc5; (iii) download an image from a link in that thread and save it as `proof_of_payment.png`; and (iv) look up the same transaction on two blockchain explorers, one of them reached from a Kraken account funding page. This sequence is **consistent with** the advertisement, communication, payment and proof-of-payment stages of the case hypothesis.

The record does **not** establish what the advertisement offered, whether the goods were controlled substances, whether any email was sent, the value or direction of the transaction, or who operated the browser. The word “Rx” in a title suggests a prescription-alternative product, which may or may not be a controlled drug. Browser-profile activity is not proof of a named person; attribution requires independent evidence such as operating-system logon records, account-provider records and device examination.

Proposition	Confidence	Basis
The profile performed the recorded sequence on 19 Apr 2022, 13:54–15:03 UTC	High	Direct records; contiguous visit IDs 1–74; no free pages (E04, E18)
The profile created and confirmed Craigslist posting 7473121658	High	Form-submission chain to confirmation and manage pages; matching posting ID in listing URL (E07–E09)
The sequence represents a sale, order and payment workflow	Moderate	Shared posting title and txid across Gmail, explorers and Kraken; content not preserved (E11–E17)
The goods advertised were controlled drugs	Low	Only the title “cheaper than Rx supplements” in category “hab”; no page content
A specific named person operated the profile	Low	History is profile-level; no independent identity evidence in scope

3.0 Evidence Acquisition and Integrity

3.1 Source of the Evidence

Field	Value
Original file name	History (no extension)
Source	ICDFA E-Campus, SBT-DF204 study material section. The course slides reference the teaching dataset published in Xu's digital-forensics-lab repository (Digital_Currency/LabFiles/History) (Xu, n.d.).
File type	SQLite 3 database (Chrome History schema version 53, last compatible version 16)
File size	196,608 bytes (41 pages × 4,096 bytes)
SHA-256 (original)	b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19
SHA-256 (working copy)	b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19 (match)
SHA-256 (post-analysis, both files)	b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19 (unchanged)
Read-only original	~/SBT-DF204-CaseStudy2/evidence/History (permissions r--r--r--)
Working copy	~/SBT-DF204-CaseStudy2/working/History_working
Tools	GNU coreutils (cp, chmod, sha256sum, ls); sqlite3 3.45.1 CLI opened with -readonly; DB Browser for SQLite (read-only mode) for visual confirmation

3.2 Working-Copy Method and Access Precautions

The supplied file was copied into an `evidence` folder and set to read-only. A second copy was made into a `working` folder, and both were hashed. The two SHA-256 values matched, establishing that the working copy is a bit-identical duplicate (Kent et al., 2006; ISO/IEC, 2012). All queries were run with `sqlite3 -readonly` so that the SQLite library could not write to the file. This matters because opening a SQLite database in normal mode can create journal or write-ahead-log files and, in some tools, update database pages. The database reports journal mode `delete`, and no `-journal` or `-wal` companion files were present before or after analysis.

```
mkdir -p ~/SBT-DF204-CaseStudy2/{evidence,working,reports,screenshots,sql}
cd ~/SBT-DF204-CaseStudy2
cp --preserve=timestamps /path/to/E-Campus/History evidence/History
chmod 444 evidence/History # original made read-only
cp evidence/History working/History_working # working copy
ls -l evidence/History
sha256sum evidence/History working/History_working | tee reports/01_acquisition.txt
# ... all analysis with: sqlite3 -readonly working/History_working ...
sha256sum evidence/History working/History_working | tee reports/99_final_hashes.txt
```

```

bash - ~/SBT-DF204-CaseStudy2 (evidence integrity)

$ ls -l evidence/History
-r--r--r-- 1 root root 196608 Oct  9 11:10 evidence/History
$ sha256sum evidence/History working/History_working
b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19  evidence/History
b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19  working/History_working

```

Figure 1. Size and SHA-256 of the read-only original and the working copy at acquisition (E01). [file: screenshots/fig01_acquisition_hash.png]

```

bash - ~/SBT-DF204-CaseStudy2 (post-analysis integrity)

$ sha256sum evidence/History working/History_working # after analysis
b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19  evidence/History
b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19  working/History_working
$ ls working/
History_working

```

Figure 2. Post-analysis SHA-256 values: both files unchanged; no journal or WAL files created (E01). [file: screenshots/fig15_final_hashes.png]

File-system timestamps on the supplied copy (owner, modification time) reflect when the file was downloaded to the analysis workstation, not activity on the suspect device. They are therefore recorded for custody purposes only and are not used as evidence of browser activity.

3.3 Chain-of-Custody Worksheet

Field	Entry
Case ID	SBT-DF204-CS2-2025-FWSD-11521
Examiner	Ibrahim Diseh Garba (2025/FWSD/11521)
Date of analysis	9 October 2026
Item	Chrome History SQLite database, 196,608 bytes
Received from	ICDFA E-Campus study material (authorised course material)
Original hash	b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19
Working-copy hash	Identical at creation and after analysis
Storage	Original read-only in evidence/; working copy in working/; SQL scripts in sql/; outputs in reports/
Handling notes	No live URL, account, wallet or transaction ID accessed. Session tokens truncated in figures.

3.4 Redaction Approach

The brief requires unrelated personal data to be redacted while retaining enough detail to verify findings. Long Google sign-in and session parameters (for example the `TL=`, `sigt=` and `osigt=` values in URL IDs 29, 31 and 32) are authentication tokens with no evidential bearing on the case and have been truncated in all figures. The Gmail address shown in page titles, the Windows profile folder `FSCS_User`, the posting ID and the transaction ID are retained because they are the identifiers on which the findings rest.

4.0 Database Method

4.1 Tables Reviewed

The schema was enumerated before any analytical query because Chrome changes its History schema across versions. The `meta` table reports schema version 53. Seventeen tables are present (Figure 3). Four carry the evidence used in this report, and two further tables were used to corroborate it.

Table	Rows	Role in this case
<code>urls</code>	57	One row per unique URL: <code>id</code> , <code>url</code> , <code>title</code> (last title seen), <code>visit_count</code> , <code>typed_count</code> , <code>last_visit_time</code> , <code>hidden</code> .
<code>visits</code>	74	One row per navigation event: <code>id</code> , <code>url</code> ($\rightarrow$ <code>urls.id</code>), <code>visit_time</code> , <code>from_visit</code> (referring visit), <code>transition</code> (bit-field), <code>visit_duration</code> (μ s), <code>opener_visit</code> (visit in the tab that opened this one).
<code>downloads</code>	1	One row per download: <code>target_path</code> , <code>current_path</code> , <code>start_time</code> , <code>end_time</code> , <code>received_bytes</code> , <code>total_bytes</code> , <code>state</code> , <code>danger_type</code> , <code>opened</code> , <code>referrer</code> , <code>tab_url</code> , <code>mime_type</code> , <code>last_modified</code> .
<code>downloads_url_chains</code>	1	Ordered URL chain for each download (<code>id</code> $\rightarrow$ <code>downloads.id</code> , <code>chain_index</code> , <code>url</code>); index 0 is the first requested URL.
<code>keyword_search_terms</code>	4	Search terms linked to the search-results URL (<code>url_id</code> $\rightarrow$ <code>urls.id</code>). Corroborates the omnibox searches.
<code>visit_source</code>	0	Records visits from non-local sources (e.g., sync or import). Empty here (see Section 7).

The joins used are `visits.url = urls.id` (each visit to its URL and title), `visits.from_visit = visits.id` (each visit to the visit that referred it) and `downloads.id = downloads_url_chains.id` (each download to its source URL chain). Schema output is reproduced in Figures 4 and 5 (Appendix 12.3).

4.2 WebKit Timestamp Conversion

Chrome stores `visit_time`, `last_visit_time`, `start_time` and `end_time` as microseconds since 1 January 1601 00:00:00 UTC. The conversion divides by 1,000,000 to obtain seconds and adds the offset between 1601-01-01 and the Unix epoch, which SQLite evaluates as `strftime('%s', '1601-01-01') = -11,644,473,600 s` (SQLite, n.d.-a). The visit timeline uses the brief's formula, which truncates to whole seconds through integer division:

```
datetime(v.visit_time/1000000 + strftime('%s', '1601-01-01'), 'unixepoch') -- whole
seconds, UTC
```

Where sub-second precision matters (the download start and end), floating-point division and an explicit offset were used. The two methods agree to the second:

```
strftime('%Y-%m-%d %H:%M:%f', d.start_time/1000000.0 - 11644473600, 'unixepoch')
-- 13294853818494344 -> 2022-04-19 14:56:58.494 UTC
```

As a cross-check, the download `start_time` was independently converted in Python (`datetime.utcfromtimestamp(13294853818494344/1e6 - 11644473600)`), giving 2022-04-19 14:56:58.494345 UTC, which matches the SQL result. `visit_duration` is also stored in microseconds and is reported in seconds ($\div 1,000,000$).

Local time. The History database does not record the device time zone. The only geographic indicator is the regional Craigslist site (`baltimore.craigslist.org`). If the device was set to US Eastern time, 19 April 2022 falls within daylight saving time, so local time = UTC – 4 h (EDT). This conversion is shown in a separate column of the timeline for reference only and is an assumption, not a finding.

4.3 Transition Interpretation

The `transition` column is a 32-bit field. The low byte (`transition & 0xFF`) is the core navigation type; the high bits are qualifiers (Chromium Authors, n.d.). SQLite returns values with bit 31 set as negative integers (e.g., `-2147483648`), so values were masked with `& 0xFFFFFFFF` before hexadecimal display. The interpretation scheme applied was:

Component	Value	Meaning used in this report
Core 0	LINK	User followed a link, or page-initiated navigation (including script-driven navigation in single-page applications)
Core 1	TYPED	URL typed into the address bar (none present in this database)
Core 5	GENERATED	Navigation generated from the omnibox, e.g., a search suggestion accepted
Core 7	FORM_SUBMIT	Result of submitting a form
0x02000000	FROM_ADDRESS_BAR	Initiated from the address bar
0x10000000	CHAIN_START	First visit in a redirect chain
0x20000000	CHAIN_END	Last visit in a redirect chain (the page actually rendered)
0x40000000	CLIENT_REDIRECT	Redirect initiated by the page (script, meta refresh, in-app routing)
0x80000000	SERVER_REDIRECT	Redirect returned by the server (HTTP 3xx)

A visit carrying both `CHAIN_START` and `CHAIN_END` (0x30000000) is a single-hop navigation. A chain such as `0x10000000 → 0x80000000 → 0xA0000000` is a link whose target was server-redirectioned twice before rendering. The transition value was never read alone: each was checked against `from_visit`, `opener_visit`, the URL and the title. Table 1 summarises every value observed.

Table 1. Transition values present in the visits table (Q15, Figure 17)

Hex value	Core	Qualifiers	n	Visit IDs
0x30000000	LINK	CHAIN_START CHAIN_END	21	2, 20, 21, 23, 34–36, 47, 52–54, 57, 58, 60–66, 71
0x80000000	LINK	SERVER_REDIRECT (mid-chain)	13	4, 12, 25–29, 32, 38–41, 69
0x10000000	LINK	CHAIN_START	12	3, 6, 9, 11, 24, 31, 37, 43, 48, 49, 55, 67
0xA0000000	LINK	SERVER_REDIRECT CHAIN_END	9	5, 7, 10, 13, 30, 33, 50, 56, 70
0x30000007	FORM_SUBMIT	CHAIN_START CHAIN_END	7	8, 14, 15, 16, 17, 18, 19
0x40000000	LINK	CLIENT_REDIRECT	5	44, 45, 68, 72, 73
0x60000000	LINK	CLIENT_REDIRECT CHAIN_END	3	42, 46, 74
0x32000005	GENERATED	FROM_ADDRESS_BAR CHAIN_START CHAIN_END	3	22, 51, 59
0x30000005	GENERATED	CHAIN_START CHAIN_END	1	1

4.4 Query Set

Fifteen saved SQL scripts (Q01–Q15) were run against the working copy and their output was written to the `reports` folder. The full text of the analytical queries is reproduced in Appendix 12.4 so that each figure can be regenerated by the examiner.

5.0 Findings

Question 1: What evidence file was acquired and how was it preserved?

The evidence item is the Chrome History SQLite database named “History”, 196,608 bytes, SHA-256 `b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19`, obtained from the SBT-DF204 study material on E-Campus. The original was made read-only, a working copy was created and hash-verified, all queries were run in read-only mode, and both files were re-hashed after analysis with identical results (Section 3; Figures 1–2; **E01**).

Question 2: Which browser-history tables and fields are relevant?

The `urls`, `visits`, `downloads` and `downloads_url_chains` tables carry the findings; `keyword_search_terms` and `visit_source` corroborate them (Section 4.1; Figures 3–5; **E02**, **E03**). Three structural observations bear on reliability. First, visit IDs run contiguously from 1 to 74 and URL IDs from 1 to 57, and the `sqlite_sequence` value for `urls` (57) equals the highest URL ID. Second, `PRAGMA freelist_count` returns 0, so the file holds no free pages in which deleted records could survive. Third, no visit precedes 13:54:18 UTC. These facts are consistent with no records having been deleted from within the window, but they do not exclude a profile that was cleared before this session, nor a database that was constructed as a whole (Figure 16; **E18**).

Question 3: When did the recorded web activity occur?

Using the conversion in Section 4.2, the first visit (ID 1) is at **2022-04-19 13:54:18 UTC** and the last (ID 74) at **2022-04-19 15:02:56 UTC**, a span of 68 min 38 s (Figures 6a–6b, 15; **E04**). Activity falls into three bursts: 13:54:18–14:05:59 (Craigslist posting, Gmail sign-in and the “order” thread); a gap of 50 min 48 s with no recorded visits; and 14:56:47–15:02:56 (transaction thread, image download, blockchain and exchange look-ups, reply compose). The gap means only that no navigation was recorded in this profile; it does not show the device was idle, since activity in another application, browser or profile would not appear here. The download record starts at 14:56:58.494 and ends at 14:57:10.261 UTC (Q10; **E14**). In assumed local time (EDT), the window is 09:54:18 – 11:02:56 on the same date.

Question 4: How did the browser reach the material URLs?

No typed navigation. Every `urls.typed_count` is 0 and no visit has core type 1 (Figure 15). The browser reached each site through omnibox search: four searches (“craigslist”, “gmail”, “blockchain explorer”, “kraken”) are recorded in `keyword_search_terms` (Figure 13; **E06**),

and their first visits (IDs 1, 22, 51, 59) carry core type 5 (GENERATED), three with the FROM_ADDRESS_BAR qualifier. Each GENERATED visit is followed within one second by a second visit to the same search URL (IDs 2, 23, 52, 60) with core type 0 and opener_visit pointing back to the first. This paired pattern appears consistently in this Chrome version; it is recorded here as an observed artefact rather than interpreted as two separate searches.

Form submissions. Seven visits have core type 7 (Table 1). Visit 8 returns to `accounts.craigslist.org/login/home` from the login page (visit 7, title “craigslist - account log in”) 17.5 s after that page loaded, which is consistent with a submitted login form. Visits 14–19 are successive form submissions through the posting pages (Question 5).

Server redirect chains. Following the Craigslist link from Google (visit 3, CHAIN_START) produced two server redirects (visit 4 `geo.craigslist.org`, 0x80000000; visit 5 `baltimore.craigslist.org`, 0xA0000000). The page rendered was therefore the Baltimore site, chosen by the server rather than by the user. Likewise, the Imgur link (visit 49) was a `google.com/url?...` wrapper that server-redirected to `imgur.com/dTgrkP7` (visit 50), and the Kraken link (visit 67) passed through a client redirect and a server redirect (68, 69) before rendering `mempool.space/tx/...` (visit 70).

Client redirects in Gmail. Gmail is a single-page application. Thread and compose views appear as LINK visits with CLIENT_REDIRECT (0x40000000 / 0x60000000) chained by `from_visit` (43→44→45→46 and 48→72→73→74). These records show changes of view inside Gmail, not separate page loads.

New tabs. Several visits have `from_visit` = 0 but a non-zero `opener_visit`. Visit 21 (public listing) has opener 20 (manage page); visit 49 (Imgur) has opener 48 (transaction thread); visit 67 (Kraken redirect to mempool) has opener 66 (Kraken funding page). The pattern is consistent with links opened in a new tab from the named page. Figure 14 shows the full `from_visit` walk for the posting chain (visit 20 back to visit 2, 19 hops) and the mempool chain (visit 70 back to 67) (E05).

Question 5: What evidence relates to the suspected sale or posting?

Table 2. Craigslist account and posting workflow (Q06, Figure 8)

Visit	UTC	Trans.	Title / URL (abridged)	Observation
6–7	13:54:22	0x10 / 0xA0	craigslist - account log in	Login page reached by redirect from account home

Visit	UTC	Trans.	Title / URL (abridged)	Observation
8	13:54:39	0x30000007	craigslist account (login/home)	Form submission from login page; account home rendered
11–13	13:55:05	link + redirects	baltimore choose type → post.craigslist.org/k/sHbOUOi.../bQTi5	New posting session token created by server redirect
14	13:55:10	0x30000007	choose category (?s=cat)	Form submitted
15	13:55:15	0x30000007	create posting (?s=edit)	Form submitted; page open 177.7 s
16	13:58:13	0x30000007	add map (?s=geoverify)	Form submitted
17	13:58:17	0x30000007	create posting (?s=editimage)	Form submitted
18	13:58:19	0x30000007	create posting (?s=preview)	Form submitted
19	13:58:22	0x30000007	baltimore posting confirmation	Form submitted; confirmation page title
20	13:58:27	0x30000000	baltimore manage posting — /manage/7473121658	Manage page for posting 7473121658
21	13:58:31	0x30000000 (opener 20)	cheaper than Rx supplements - health and beauty - by owner — baltimore.craigslist.org/hab/d/.../7473121658.html	Public listing for the same posting ID, opened from the manage page

Supporting the hypothesis. The workflow is complete and ordered: category selection, editing, map, image, preview, confirmation and management, each reached by form submission from the previous step through an unbroken `from_visit` chain (Figure 14). The posting ID 7473121658 appears in both the management URL (URL ID 15) and the public listing URL (URL ID 16), linking the posting session to a published listing. The 177.7 s duration on the “create posting” page (visit 15) is the longest in the posting chain and is consistent with composing listing text, although duration measures time the page was open, not typing (E07–E09).

Challenging or limiting the hypothesis. The listing title reads “cheaper than Rx supplements” and the URL places it in `hab` (health and beauty), posted “by owner”. Neither the History file nor the title identifies a controlled substance. “Rx supplements” could describe lawful supplements marketed as alternatives to prescription products. The `?s=editimage` step shows the image-editing page was reached, not that an image was uploaded. The advertisement text is not in the database, and whether the listing stayed live or was flagged cannot be determined.

Question 6: What evidence relates to communications and payment activity?

6.1 Email (Gmail)

After an omnibox search for “gmail” (visit 22), the browser passed through the Google sign-in pages, including the password-challenge URL (visit 36, 13:59:26), and reached the inbox at 13:59:29–13:59:31 (visits 40–42). Page titles from visit 25 onward read “Inbox (12) - unsub.fsccs@gmail.com - Gmail”, identifying the account displayed (E10). The password-

challenge visit indicates an interactive sign-in from this browser rather than a pre-existing session, but the History file does not record whether the password was typed or auto-filled.

Table 3. Case-relevant Gmail thread views (Q07, Figure 9)

Visit	UTC	Thread (ID prefix)	Title (abridged)	View
43	14:04:12	A: WhctKKXXDrfCqXBG	cheaper than Rx supplements - order	Thread opened from inbox
44, 45	14:04:35, 14:04:46	A	same	Two compose states (distinct compose tokens)
46, 47	14:04:52, 14:05:59	A → inbox	same / Inbox (12)	Return to thread, then inbox
48	14:56:47	B: WhctKKXXDrfCqvtR	cheaper than Rx supplements -- txid: 517b21569149...	Thread opened from inbox
72, 73	15:02:32, 15:02:44	B	cheaper than Rx supplements - order	Two compose states
74	15:02:56	B	cheaper than Rx supplements -- txid: ...	Return to thread; last visit

Thread A’s title ties an “order” to the posting title; Thread B’s title carries the full 64-character transaction ID that appears later in the explorer and Kraken records (**E11**, **E12**). The compose-state URLs show that a reply or new-message window was opened in each thread. They do not show that a message was written or sent; a sent message would need confirmation from the Gmail account or the provider. The `urls.title` field holds only the last title seen for a URL, and Gmail updates titles dynamically. That explains why compose URLs in Thread B (URL IDs 56–57) carry the “- order” title while the thread URL (ID 37) carries the txid title; the titles show the label displayed at capture, not necessarily the subject of the message being composed.

6.2 Image hosting and the proof-of-payment download

Five seconds after Thread B opened, a link in it was followed in a new tab (visit 49, opener 48). The URL `google.com/url?q=https://imgur.com/dTgrkP7&source=gmail...` is Gmail’s link-wrapping format, showing that the link came from email content rather than the address bar. It server-redirected to `imgur.com/dTgrkP7` (visit 50, page open 18.9 s). The Gmail wrapper also contains a `ust` parameter (1650466604648000) that decodes to 2022-04-20 14:56:44 UTC, exactly 24 hours after a moment three seconds before visit 48. Google does not document this parameter, so it is noted as corroboration only (**E13**).

Table 4. Download record (Q10, Figure 12)

Field	Value	Interpretation
<code>downloads.id</code>	1	Only download in the profile
<code>target_path / current_path</code>	<code>C:\Users\FSCS_User\Desktop\proof_of_payment.png</code>	Saved to the Desktop of Windows user folder FSCS_User
<code>downloads_url_chains (index 0)</code>	<code>https://i.imgur.com/dTgrkP7.png</code>	Direct image URL; same Imgur ID as visit 50

Field	Value	Interpretation
tab_url / referrer	https://imgur.com/dTgrkP7 / https://imgur.com/	Download initiated from the Imgur page tab
start_time → end_time (UTC)	14:56:58.494 → 14:57:10.261	6.5 s after the Imgur page loaded; 11.8 s duration
received / total bytes	33,844 / 33,844	Complete transfer
state / danger_type	1 (complete) / 0 (not dangerous)	Download finished without warning
opened	0	Not opened through Chrome (opening via file explorer would not be recorded)
mime_type / last_modified	image/png / Tue, 19 Apr 2022 14:52:06 GMT	Server reports image last modified 4 min 41 s before Thread B was viewed

The saved name `proof_of_payment.png` differs from the server name `dTgrkP7.png`. A name chosen at save time (for example, through “Save image as”) would produce this difference, which suggests deliberate labelling of the file. A server-supplied file name could in principle produce the same result, so this is an inference. The file itself is not in the History database; its existence, hash and content would need to be confirmed from the device’s file system (E14).

6.3 Blockchain explorers and exchange

Table 5. Transaction ID correlation across sites (Q09, Figure 11)

Visit	UTC	Site	How reached	Record
48	14:56:47	Gmail	Thread opened from inbox	txid in page title only
55–57	14:59:06	blockchain.com	Search from explorer page (54) → server redirect	search?search=<txid> → /btc/tx/<txid>
58	14:59:21	blockchain.com	From tx page (opener 57)	Address 38RcsURWCDmCbYocmsZCnGz1FtD7x477mt
61–63	14:59:47– 15:00:21	kraken.com	From Google search (60)	Sign-in; “Approve new device” page open 104.4 s
64–66	15:01:11– 15:01:21	kraken.com	From device approval (63)	trade, instant, funding pages
67–70	15:01:31– 15:01:33	kraken.com → mempool.space	New tab from funding page (opener 66); client + server redirect	mempool.space/tx/<txid>
71	15:02:05	kraken.com	From device approval chain (63)	History – Ledger page

The same transaction ID appears in a Gmail thread title, in a blockchain.com search typed into the explorer’s search field (the `search?search=` URL shows the query value), and in a Kraken outbound link to mempool.space opened from the Kraken funding page (E15–E17). The last point is the strongest payment link in the database: a redirect URL generated by Kraken from the funding page is consistent with the transaction being listed in the signed-in Kraken account’s funding history, such as a deposit. The History record does not show the direction, amount, confirmation status or the Kraken account identity.

The address visited on blockchain.com (visit 58) was reached from the transaction page, which suggests it is an input or output of that transaction, but its role and owner are not recorded. Two Kraken titles contain US-dollar figures (“41,543.80 USD” at 15:01:21 and “41,563.70 USD” at 15:02:05). Because the figure changed within 44 seconds, it may be a live market price shown in the page title rather than an account balance. It is not treated as a financial finding. The “Approve new device” page indicates that Kraken did not recognise this browser or device at sign-in, which may mean the account was normally used elsewhere; this requires confirmation from Kraken records.

Question 7: What is the reconstructed timeline?

The timeline is presented in Section 6 (summary) and Appendix 12.2 (complete, with record IDs and assumed local time). It separates the observed artefact from the interpretation and its limits (E19).

Question 8: What conclusion can be defended?

Direct observations. On 19 April 2022, between 13:54:18 and 15:02:56 UTC, this Chrome profile (a) completed a Craigslist posting workflow ending in posting 7473121658 titled “cheaper than Rx supplements”; (b) displayed Gmail account unsub.fscs@gmail.com and two threads titled with that posting title, the word “order” and a Bitcoin transaction ID; (c) followed a link from the transaction thread to an Imgur image and saved it as proof_of_payment.png; and (d) displayed that transaction on blockchain.com and, from a Kraken funding page, on mempool.space.

Reasonable inferences. The order of these events (posting, order thread, transaction thread, proof image, independent transaction checks, exchange funding review, reply compose) fits the case hypothesis of advertising, buyer contact, payment notification and verification. The posting being created in this profile, and the transaction being checked from the exchange funding page, fit a **seller** receiving payment better than a buyer making it, though the record does not exclude the reverse.

Not supported by the History database alone. That the goods were controlled drugs; the content of the listing, emails or image; that any message was sent; that payment was made for the posting; the value of the transaction; and the identity of the operator. Section 7 sets out limitations and alternative explanations.

6.0 Timeline (Summary, UTC)

Table 6 condenses the 74 visits and one download into sixteen activity stages. The complete timeline, with every material record ID and an assumed-local-time column, is in Appendix 12.2.

Table 6. Summary of reconstructed activity stages

#	UTC (19 Apr 2022)	Records	Stage	Observed
1	13:54:18–13:54:19	V1–5	Reach Craigslist	Omnibox search; link redirected to Baltimore site
2	13:54:22–13:54:39	V6–8	Account sign-in	Login page; form submission to account home
3	13:55:03–13:58:22	V9–19	Create posting	Six form submissions through posting steps to confirmation
4	13:58:27–13:58:31	V20–21	Manage / view posting	Posting 7473121658; listing “cheaper than Rx supplements” (hab)
5	13:58:38–13:59:31	V22–42	Gmail sign-in	Password challenge; inbox of unsub.fscc@gmail.com
6	14:04:12–14:05:59	V43–47	“Order” thread	Thread A viewed; two compose states; back to inbox
—	14:05:59–14:56:47	—	No recorded activity	Gap of 50 min 48 s
7	14:56:47	V48	Transaction thread	Thread B title contains txid 517b...afdc5
8	14:56:52	V49–50	Image link	Gmail-wrapped link → imgur.com/dTrkP7 (new tab)
9	14:56:58–14:57:10	D1	Download	proof_of_payment.png, 33,844 bytes, complete
10	14:58:59–14:59:06	V51–57	Explorer look-up	blockchain.com search for txid → tx page
11	14:59:21	V58	Address view	38RcsURWCDmCbYocmsZCnGz1FtD7x477mt
12	14:59:38–15:00:21	V59–63	Exchange sign-in	Kraken sign-in; new-device approval
13	15:01:11–15:01:21	V64–66	Exchange account	Trade, instant, funding pages
14	15:01:31–15:01:33	V67–70	Second explorer	From funding page via Kraken redirect → mempool.space/tx/<txid>
15	15:02:05	V71	Ledger	Kraken History – Ledger page
16	15:02:32–15:02:56	V72–74	Reply compose	Two compose states in Thread B; last visit

```

sqlite3 -readonly working/History_working
sqlite> .read sql/q09_txid.sql
vid  uid  visit_utc      in_url  in_title  host_path
-----
48   37   2022-04-19 14:56:47 -       Y        mail.google.com/mail/u/0/#inbox/Wh
55   42   2022-04-19 14:59:06 Y       Y        www.blockchain.com/search?search=5
56   43   2022-04-19 14:59:06 Y       Y        www.blockchain.com/btc/tx/517b2156
57   43   2022-04-19 14:59:06 Y       Y        www.blockchain.com/btc/tx/517b2156
67   51   2022-04-19 15:01:31 Y       Y        www.kraken.com/redirect?url=https%
68   52   2022-04-19 15:01:33 -       Y        ww.kraken.com/redirect?t=-_F22Ihws
69   53   2022-04-19 15:01:33 -       Y        www.kraken.com/redirect?t=-_F22Ihw
70   54   2022-04-19 15:01:33 Y       Y        mempool.space/tx/517b2156914944339
74   37   2022-04-19 15:02:56 -       Y        mail.google.com/mail/u/0/#inbox/Wh

```

Figure 11 (repeated from Appendix 12.3). Every visit whose URL or title contains the transaction ID (E15–E17). [file: screenshots/fig10_txid_correlation.png]

7.0 Attribution Assessment

Interpretation follows the principle that digital evidence should be evaluated against competing explanations and reported with stated limits (ISO/IEC, 2015; Forensic Science Regulator, 2023). Confidence is graded as **High** (directly recorded, internally consistent, no credible competing explanation within the data), **Moderate** (inference supported by several correlated records but dependent on content or context not preserved) or **Low** (not established by this evidence; requires independent sources). Numerical probabilities are not assigned because the History database offers no basis for calibrating them.

Table 7. Observation, inference and confidence

Category	Content
Direct observations	URLs, titles, UTC visit times, transition values, from_visit/opener links, search terms, one completed download with source URL and saved path.
Reasonable inferences	One profile carried out a posting → order → payment-notification → verification workflow in a 68-minute window; the profile's user saved the image under a descriptive name; the transaction was visible in a Kraken account's funding view.
Confidence (profile activity)	High
Confidence (sale/payment workflow)	Moderate
Confidence (controlled drugs)	Low
Confidence (human identity)	Low
Would raise confidence	Windows logon records for FSCS_User at the times recorded; Craigslist records for posting 7473121658; Gmail content for threads A and B; Kraken account and deposit records for the txid; recovery and hashing of proof_of_payment.png; Chrome Login Data, Web Data (autofill) and Sessions artefacts from the same profile.
Would lower confidence	Evidence of a shared Windows account or Chrome profile; remote-access tools active during the window; synced or imported history; automation or extension activity; evidence the dataset was modified after collection.

7.1 Limitations and Alternative Explanations

- L1. **Browser profile is not a person.** History records what a Chrome profile under Windows user folder FSCS_User did. Anyone with access to that logged-in account could have produced it. The Gmail address and Kraken session show which accounts were signed in, not who typed the credentials.
- L2. **Missing content.** The database stores titles and URLs, not page bodies. The listing text, email messages, image content and exchange data are absent. A title containing “Rx” or “order” is not proof of what was offered or agreed.
- L3. **Redirects and dynamic titles.** Several visits are server or client redirects that the user did not choose (for example, `geo.craigslist.org` → `baltimore.craigslist.org`).

Gmail titles change dynamically, and `urls.title` keeps only the last title, so a title may not match the view at the moment of each visit.

- L4. **Cached or background loads.** A visit record means navigation occurred in the browser. Page elements may have been served from cache, and short durations (several visits record 0 s) do not show that a person read the page.
- L5. **Synced, remote or automated activity.** The `visit_source` table is empty. Chrome uses that table to mark visits that came from sources other than local browsing, such as sync, so its emptiness is consistent with local browsing on this device; this reading should be verified against the Chromium source for schema version 53. Remote-desktop sessions or automation would still appear as local browsing and cannot be excluded from this file.
- L6. **Role ambiguity.** The profile could belong to a buyer verifying their own payment rather than a seller confirming receipt. Creation of the posting in the same profile favours the seller reading but does not prove it.
- L7. **Lawful explanation.** The listing could relate to lawful supplements, with the cryptocurrency payment entirely legitimate. Nothing in the database distinguishes these scenarios.
- L8. **Dataset provenance.** The file is authorised course material. The repository that publishes it also contains files named “Chrome_History_Recreation_...”, which indicates that the scenario was staged for teaching. Findings are reported as they would be for genuine casework, but provenance of a real exhibit would need to be established through device seizure records.

8.0 Challenges Encountered

Challenge	Resolution
Negative transition values in SQLite (e.g., -2147483648)	Masked with <code>transition & 0xFFFFFFFF</code> before hexadecimal display; decoded qualifiers bit by bit (Q05).
Paired GENERATED/LINK visits for each omnibox search	Recorded as an observed artefact, linked by <code>opener_visit</code> ; not counted as separate user searches.
Visits with <code>from_visit = 0</code> that clearly continued a workflow	Used <code>opener_visit</code> to identify pages opened in new tabs (visits 21, 49, 67).
Gmail single-page application	Interpreted <code>CLIENT_REDIRECT</code> chains as view changes; distinguished threads by URL thread-ID prefix rather than title.
Integer division truncates sub-second time	Used floating-point division for the download record; cross-checked in Python.
Very long authentication URLs in figures	Truncated output columns; retained only case-relevant identifiers.
No device time zone in History	Kept UTC as the reference; local time shown separately and labelled as an assumption.

9.0 Recommendations

The following recommendations concern next investigative steps and examination practice. They are distinct from the findings above.

9.1 Further Investigative Steps

Action	Purpose
Examine the full Chrome profile (Login Data, Web Data, Cookies, Sessions, Cache) and the Windows image of FSCS_User	Recover page content, autofill and session state; locate and hash <code>proof_of_payment.png</code>
Correlate Windows Security event logs (logon/logoff) with 13:54–15:03 UTC on 19 Apr 2022	Link the browser activity to an authenticated user session
Seek, through lawful process, Craigslist records for posting 7473121658	Obtain advertisement text, images and poster account details
Seek Google records for unsub.fscs@gmail.com (threads A and B)	Establish message content, sender, recipient and whether replies were sent
Seek Kraken records for the account signed in at 14:59–15:02 UTC	Confirm whether the txid was a deposit to that account, its value and the account holder (KYC)
Trace the transaction and address 38Res... with blockchain analytics (offline, authorised tools)	Establish the flow of funds without interacting with live services from the case environment

9.2 Examination Practice

- Open SQLite evidence in read-only mode and hash before and after analysis, because routine opening can create journal files or modify pages.
- Enumerate the schema and `meta.version` before querying; field sets differ between Chrome versions (for example, `opener_visit` is absent in older schemas).
- Decode the full transition bit-field and read it with `from_visit`, `opener_visit`, URLs and titles; a core type alone can mislead in single-page applications.
- Report UTC as the reference time and label any local-time conversion as an assumption unless the device time zone is established from the operating system.
- Corroborate History findings with a second tool (for example, a browser-artefact parser such as Hindsight) to guard against single-tool error (Obsidian Forensics, n.d.).

10.0 Conclusion

The examination preserved the History database with matching SHA-256 values before and after analysis, identified the relevant Chrome tables, converted WebKit timestamps to UTC and decoded navigation transitions. From these, a 68-minute sequence on 19 April 2022 was reconstructed in which one Chrome profile created a Craigslist posting titled “cheaper than Rx supplements”, viewed Gmail threads linking that title to an order and to a Bitcoin transaction, saved an image named proof_of_payment.png from a link in that thread, and checked the same transaction on two blockchain explorers, one opened from a Kraken funding page.

Two shared identifiers, posting ID 7473121658 and transaction 517b2156...afdc5, bind the records together across five services, and the navigation metadata shows how each page was reached. That coherence supports the case hypothesis at a moderate level for the sale-and-payment workflow. The evidence does not establish that the goods were controlled drugs or who operated the browser. Those questions depend on page content, provider records and device-level attribution evidence that lie outside this database.

11.0 References

- Casey, E. (2011). *Digital evidence and computer crime: Forensic science, computers and the Internet* (3rd ed.). Academic Press.
- Chromium Authors. (n.d.). *page_transition_types.h* [Source code]. Chromium. Retrieved October 9, 2026, from https://source.chromium.org/chromium/chromium/src/+/main:ui/base/page_transition_types.h
- Forensic Science Regulator. (2023). *Forensic science regulator code of practice* (Version 1). UK Government. [Verify edition and URL before submission.]
- International Cybersecurity and Digital Forensics Academy. (2026a). *SBT-DF204 Computer forensics case studies: Case study 2 student assessment brief* [Unpublished course material]. ICDFA E-Campus.
- International Cybersecurity and Digital Forensics Academy. (2026b). *Understand URL and user web activity* [Lecture slides]. ICDFA E-Campus.
- International Organization for Standardization & International Electrotechnical Commission. (2012). *Information technology — Security techniques — Guidelines for identification, collection, acquisition and preservation of digital evidence* (ISO/IEC 27037:2012).
- International Organization for Standardization & International Electrotechnical Commission. (2015). *Information technology — Security techniques — Guidelines for the analysis and interpretation of digital evidence* (ISO/IEC 27042:2015).
- Kent, K., Chevalier, S., Grance, T., & Dang, H. (2006). *Guide to integrating forensic techniques into incident response* (NIST Special Publication 800-86). National Institute of Standards and Technology. <https://doi.org/10.6028/NIST.SP.800-86>
- Obsidian Forensics. (n.d.). *Hindsight: Web browser forensics for Google Chrome/Chromium* [Computer software]. GitHub. <https://github.com/obsidianforensics/hindsight>
- SQLite. (n.d.-a). *Date and time functions*. Retrieved October 9, 2026, from https://www.sqlite.org/lang_datefunc.html

SQLite. (n.d.-b). *Command line shell for SQLite*. Retrieved October 9, 2026, from <https://www.sqlite.org/cli.html>

Xu, F. (n.d.). *digital-forensics-lab* [GitHub repository]. GitHub. Retrieved October 9, 2026, from <https://github.com/frankwxu/digital-forensics-lab>

12.0 Appendices

12.1 Evidence Log

V = visits.id; URL = urls.id. Query numbers refer to Appendix 12.4; figure numbers to Appendix 12.3.

ID	Source record	Finding	Why it matters	SQL / figure
E01	evidence/History; working/History_working	196,608 bytes; SHA-256 b991b0fa...3fd19 identical at acquisition and after analysis	Integrity of the exhibit and working copy	Fig. 1, 2
E02	sqlite_master; meta	17 tables; schema version 53	Confirms field set used by queries	Q01–Q02; Fig. 3
E03	PRAGMA table_info (urls, visits, downloads, downloads_url_chains)	Fields incl. from_visit, transition, opener_visit, visit_duration	Basis for joins and navigation analysis	Q03; Fig. 4, 5
E04	visits ⇄ urls, V1–V74	Window 13:54:18–15:02:56 UTC on 19 Apr 2022; gap 14:05:59–14:56:47	Defines timeline and activity bursts	Q04, Q13; Fig. 6a, 6b, 15
E05	visits.transition, from_visit, opener_visit	9 distinct transition values; redirect and new-tab chains decoded	Shows how material URLs were reached	Q05, Q12, Q15; Fig. 7a, 7b, 14, 17
E06	keyword_search_terms (url_id 1, 17, 40, 45); urls.typed_count	Searches: craigslist, gmail, blockchain explorer, kraken; typed_count = 0 for all URLs	Navigation by omnibox search, not typed URLs	Q11, Q13; Fig. 13, 15
E07	V6–V8 (URL 5, 6)	Craigslist login page, then FORM_SUBMIT (0x30000007) to account home at 13:54:39	Account sign-in before posting	Q06; Fig. 8
E08	V11–V19 (URL 7–14, 8)	Posting session bQTi5; form submissions through cat, edit (177.7 s), geoverify, editimage, preview, confirmation	Posting workflow completed in this profile	Q06, Q12; Fig. 8, 14
E09	V20 (URL 15); V21 (URL 16)	manage/7473121658; listing “cheaper than Rx supplements”, hab, same ID; opener 20	Links posting session to a published listing	Q06; Fig. 8
E10	V22–V42 (URL 17–33)	Gmail sign-in incl. password challenge (V36); titles show unsub.fscs@gmail.com	Identifies email account displayed	Q05, Q07; Fig. 7a, 9
E11	V43–V47 (URL 33–36)	Thread A “cheaper than Rx supplements - order”; two compose states	Order communication tied to posting title	Q07; Fig. 9
E12	V48, V72–V74 (URL 37, 56, 57)	Thread B title contains txid; compose states at 15:02:32–15:02:44	Payment notification and later reply activity	Q07, Q09; Fig. 9, 11
E13	V49–V50 (URL 38, 39)	Gmail-wrapped link (opener 48) → imgur.com/dTgrkP7	Image reached from transaction email	Q08; Fig. 10
E14	downloads.id 1; downloads_url_chains id 1	proof_of_payment.png, 33,844 B, complete, from i.imgur.com/dTgrkP7.png, 14:56:58.494 UTC	Retained proof-of-payment artefact	Q10; Fig. 12
E15	V51–V58 (URL 40–44)	blockchain.com search and tx page for txid; address 38Rcs... viewed	Independent transaction verification	Q08, Q09; Fig. 10, 11
E16	V59–V66, V71 (URL 45–50, 55)	Kraken sign-in, new-device approval, trade, instant, funding, ledger	Exchange account activity in window	Q08; Fig. 10
E17	V67–V70 (URL 51–54)	From funding page (opener 66) via Kraken redirect to mempool.space/tx/<txid>	Ties txid to exchange account view	Q08, Q09, Q12; Fig. 10, 11, 14
E18	visits/urls IDs; sqlite_sequence; freelist	IDs contiguous 1–74 and 1–57; urls seq 57; freelist 0	No sign of deleted records within window	Q14; Fig. 16
E19	Compilation of E04–E17	UTC timeline of 16 stages	Chronological account (Question 7)	Section 6; Appendix 12.2

12.2 Timeline Table (UTC)

All times 19 April 2022. UTC is the reference. The local column assumes US Eastern Daylight Time (UTC – 4 h), inferred only from the Baltimore regional site, and is not a finding. V = visits.id, U = urls.id, D = downloads.id.

Stage	UTC timestamp	Local (EDT, assumed)	Visit / URL ID	Artefact	Observed fact	Interpretation and limit
1. Reach site	13:54:18	09:54:18	V1 / U1	google.com/search?q=craigslist	GENERATED 0x30000005; search term “craigslist”	Omnibox search; who typed it is unknown
1. Reach site	13:54:19	09:54:19	V3–V5 / U2–U4	craigslist.org → geo → baltimore	Link, then two server redirects (0x80000000, 0xA0000000)	Region chosen by server, not user
2. Sign-in	13:54:22	09:54:22	V6–V7 / U5–U6	accounts.craigslist.org/login	Redirect to “account log in” page; open 17.5 s	Login page displayed
2. Sign-in	13:54:39	09:54:39	V8 / U5	accounts.craigslist.org/login/home	FORM_SUBMIT 0x30000007 from V7	Consistent with login form submitted; account identity not recorded
3. Posting	13:55:05	09:55:05	V11–V13 / U7–U9	post.craigslist.org/c/bal → /k/.../bQTi5	Server redirect creates posting session	Start of posting workflow
3. Posting	13:55:10–13:55:15	09:55:10–09:55:15	V14–V15 / U10–U11	?s=cat, ?s=edit	FORM_SUBMIT; edit page open 177.7 s	Category chosen; listing composed (text not recorded)
3. Posting	13:58:13–13:58:19	09:58:13–09:58:19	V16–V18 / U12–U14	?s=geoverify, editimage, preview	FORM_SUBMIT ×3	Map, image step, preview; image upload not proven
3. Posting	13:58:22	09:58:22	V19 / U8	posting confirmation	FORM_SUBMIT	Posting submitted for publication
4. Manage / view	13:58:27	09:58:27	V20 / U15	post.craigslist.org/manage/7473121658	LINK from V19	Posting ID 7473121658 assigned
4. Manage / view	13:58:31	09:58:31	V21 / U16	baltimore.craigslist.org/hab/d/.../7473121658.html	Title “cheaper than Rx supplements - health and beauty - by owner”; opener V20	Public listing viewed; content and lawfulness not recorded
5. Email sign-in	13:58:38–13:59:26	09:58:38–09:59:26	V22–V36 / U17–U29	google search “gmail” → accounts.google.com	Redirect chains; password-challenge page V36	Interactive Google sign-in in this browser
5. Email sign-in	13:59:29–13:59:31	09:59:29–09:59:31	V37–V42 / U30–U33, U19–U20	mail.google.com/mail/u/0/#inbox	Title “Inbox (12) - unsub.fscs@gmail.com”	Account displayed; operator unknown
6. Order thread	14:04:12	10:04:12	V43 / U34	Gmail thread A	Title “cheaper than Rx supplements - order”	Order-related email viewed; body not recorded
6. Order thread	14:04:35–14:04:52	10:04:35–10:04:52	V44–V46 / U35, U36, U34	thread A ?compose=...	Two compose states (CLIENT_REDIRECT)	Reply window opened; sending not proven
6. Order thread	14:05:59	10:05:59	V47 / U33	inbox	Return to inbox	End of first burst
— Gap	14:05:59–14:56:47	10:05:59–10:56:47	—	—	No visits for 50 min 48 s	Inactivity in this profile only
7. Txid thread	14:56:47	10:56:47	V48 / U37	Gmail thread B	Title “cheaper than Rx supplements -- txid: 517b...afdc5”	Payment notification consistent with title; content not recorded

Stage	UTC timestamp	Local (EDT, assumed)	Visit / URL ID	Artefact	Observed fact	Interpretation and limit
8. Image link	14:56:52	10:56:52	V49–V50 / U38–U39	google.com/url?q=imgur... → imgur.com/dTgrkP7	Opener V48; server redirect; open 18.9 s	Link followed from email content
9. Download	14:56:58.494–14:57:10.261	10:56:58–10:57:10	D1 (chain idx 0)	C:\Users\FSCS_User\Desktop\proof_of_payment.png	33,844 B complete; from i.imgur.com/dTgrkP7.png; opened = 0	Proof image saved under descriptive name; file content unverified
10. Explorer	14:58:59–14:59:02	10:58:59–10:59:02	V51–V54 / U40–U41	google search “blockchain explorer” → blockchain.com/explorer	GENERATED + FROM_ADDRESS_BAR	Explorer opened
10. Explorer	14:59:06	10:59:06	V55–V57 / U42–U43	blockchain.com/search?search=<txid> → /btc/tx/<txid>	Search query in URL; server redirect	Txid entered in explorer; result not recorded
11. Address	14:59:21	10:59:21	V58 / U44	blockchain.com/btc/address/38Rcs...477mt	Opener V57 (tx page)	Address linked to tx; role/owner unknown
12. Exchange sign-in	14:59:38–15:00:21	10:59:38–11:00:21	V59–V63 / U45–U47	kraken.com sign-in → device-approval	Device-approval page open 104.4 s	Sign-in from unrecognised device; account identity not recorded
13. Exchange account	15:01:11–15:01:21	11:01:11–11:01:21	V64–V66 / U48–U50	kraken.com/u/trade, /instant, /funding	Funding title “41,543.80 USD ...”	Account pages viewed; figure may be market price, not balance
14. Second explorer	15:01:31–15:01:33	11:01:31–11:01:33	V67–V70 / U51–U54	kraken.com/redirect → mempool.space/tx/<txid>	Opener V66 (funding); client + server redirect	Txid shown in Kraken funding context; direction/amount unknown
15. Ledger	15:02:05	11:02:05	V71 / U55	kraken.com/u/history/ledger	Title “41,563.70 USD ...”	Ledger viewed; entries not recorded
16. Reply	15:02:32–15:02:56	11:02:32–11:02:56	V72–V74 / U56, U57, U37	Gmail thread B ?compose=...	Two compose states; return to thread (last visit)	Reply drafted after verification; sending not proven

12.3 Labelled SQL Output Screenshots

Each figure is the console output of the named saved query run with `sqlite3 -readonly working/History_working`. Long URLs and titles are truncated by the query itself; authentication tokens are not displayed. Figures 1, 2 and 11 also appear in the body of the report.

```

sqlite3 -readonly working/History_working

sqlite> SELECT name FROM sqlite_master WHERE type='table' ORDER BY name;
clusters
clusters_and_visits
content_annotations
context_annotations
downloads
downloads_reroute_info
downloads_slices
downloads_url_chains
keyword_search_terms
meta
segment_usage
segments
sqlite_sequence
typed_url_sync_metadata
urls
visit_source
visits

sqlite> SELECT key, value FROM meta;
key                                value
-----
mmap_status                        -1
last_compatible_version            16
version                            53
early_expiration_threshold         13287337139605076

sqlite> -- row counts of tables used
tbl                                n
-----
urls                               57
visits                             74
downloads                          1
downloads_url_chains               1
keyword_search_terms               4
visit_source                       0

```

Figure 3. Table list, meta values (schema version 53) and row counts (Q01–Q02; E02). [file: screenshots/fig02_tables_meta.png]

```

sqlite3 -readonly working/History_working

sqlite> PRAGMA table_info(urls);
cid  name      type      notnull  dflt_value  pk
---  -
0    id        INTEGER   0        0            1
1    url       LONGVARCHAR 0        0            0
2    title     LONGVARCHAR 0        0            0
3    visit_count INTEGER   1        0            0
4    typed_count INTEGER   1        0            0
5    last_visit_time INTEGER  1        0            0
6    hidden    INTEGER   1        0            0

sqlite> PRAGMA table_info(visits);
cid  name      type      notnull  dflt_value  pk
---  -
0    id        INTEGER   0        0            1
1    url       INTEGER   1        0            0
2    visit_time INTEGER   1        0            0
3    from_visit INTEGER   0        0            0
4    transition INTEGER   1        0            0
5    segment_id INTEGER   0        0            0
6    visit_duration INTEGER  1        0            0
7    incremented_omnibox_typed_score BOOLEAN  1        FALSE        0
8    opener_visit INTEGER   0        0            0

sqlite> PRAGMA table_info(downloads_url_chains);
cid  name      type      notnull  dflt_value  pk
---  -
0    id        INTEGER   1        0            1
1    chain_index INTEGER   1        0            2
2    url       LONGVARCHAR 1        0            0

```

Figure 4. PRAGMA table_info for urls, visits and downloads_url_chains (Q03; E03). [file: screenshots/fig03_schema_urls_visits.png]

```

sqlite3 -readonly working/History_working
sqlite> PRAGMA table_info(downloads);
cid  name                                type                notnull  dflt_value  pk
---  -
0     id                                  INTEGER            0         0           1
1     guid                               VARCHAR            1         0           0
2     current_path                       LONGVARCHAR        1         0           0
3     target_path                       LONGVARCHAR        1         0           0
4     start_time                         INTEGER            1         0           0
5     received_bytes                     INTEGER            1         0           0
6     total_bytes                       INTEGER            1         0           0
7     state                             INTEGER            1         0           0
8     danger_type                       INTEGER            1         0           0
9     interrupt_reason                  INTEGER            1         0           0
10    hash                              BLOB               1         0           0
11    end_time                         INTEGER            1         0           0
12    opened                          INTEGER            1         0           0
13    last_access_time                 INTEGER            1         0           0
14    transient                        INTEGER            1         0           0
15    referrer                         VARCHAR            1         0           0
16    site_url                         VARCHAR            1         0           0
17    tab_url                         VARCHAR            1         0           0
18    tab_referrer_url                 VARCHAR            1         0           0
19    http_method                     VARCHAR            1         0           0
20    by_ext_id                       VARCHAR            1         0           0
21    by_ext_name                     VARCHAR            1         0           0
22    etag                            VARCHAR            1         0           0
23    last_modified                   VARCHAR            1         0           0
24    mime_type                       VARCHAR(255)       1         0           0
25    original_mime_type              VARCHAR(255)       1         0           0
26    embedder_download_data          VARCHAR            1         ''          0

```

Figure 5. PRAGMA table_info for downloads (Q03; E03). [file: screenshots/fig04_schema_downloads.png]

sqlite3 -readonly working/History_working (visits 1-37)

```
sqlite> .read sql/q04_timeline.sql
```

vid	uid	visit_utc	from_v	opener	trans_hex	core	dur_s	title
1	1	2022-04-19 13:54:18	0	0	0x30000005	5	0.0	craigslist - Google Search
2	1	2022-04-19 13:54:19	0	1	0x30000000	0	0.7	craigslist - Google Search
3	2	2022-04-19 13:54:19	2	0	0x10000000	0	0.0	craigslist: baltimore jobs, apartments, for
4	3	2022-04-19 13:54:19	3	0	0x80000000	0	0.0	craigslist: baltimore jobs, apartments, for
5	4	2022-04-19 13:54:19	4	0	0xA0000000	0	2.5	craigslist: baltimore jobs, apartments, for
6	5	2022-04-19 13:54:22	5	0	0x10000000	0	0.0	craigslist account
7	6	2022-04-19 13:54:22	6	0	0xA0000000	0	17.5	craigslist - account log in
8	5	2022-04-19 13:54:39	7	0	0x30000007	7	23.4	craigslist account
9	2	2022-04-19 13:55:03	8	0	0x10000000	0	0.0	craigslist: baltimore jobs, apartments, for
10	4	2022-04-19 13:55:03	9	0	0xA0000000	0	2.7	craigslist: baltimore jobs, apartments, for
11	7	2022-04-19 13:55:05	10	0	0x10000000	0	0.0	baltimore choose type
12	8	2022-04-19 13:55:05	11	0	0x80000000	0	0.0	baltimore posting confirmation
13	9	2022-04-19 13:55:05	12	0	0xA0000000	0	4.3	baltimore choose type
14	10	2022-04-19 13:55:10	13	0	0x30000007	7	5.6	baltimore choose category
15	11	2022-04-19 13:55:15	14	0	0x30000007	7	177.7	baltimore create posting
16	12	2022-04-19 13:58:13	15	0	0x30000007	7	4.4	baltimore add map
17	13	2022-04-19 13:58:17	16	0	0x30000007	7	1.6	baltimore create posting
18	14	2022-04-19 13:58:19	17	0	0x30000007	7	3.2	baltimore create posting
19	8	2022-04-19 13:58:22	18	0	0x30000007	7	5.0	baltimore posting confirmation
20	15	2022-04-19 13:58:27	19	0	0x30000000	0	5.6	baltimore manage posting
21	16	2022-04-19 13:58:31	0	20	0x30000000	0	0.0	cheaper than Rx supplements - health and bea
22	17	2022-04-19 13:58:38	0	0	0x32000005	5	0.0	gmail - Google Search
23	17	2022-04-19 13:58:39	0	22	0x30000000	0	0.8	gmail - Google Search
24	18	2022-04-19 13:58:40	23	0	0x10000000	0	0.0	Gmail: Free, Private & Secure Email Google
25	19	2022-04-19 13:58:40	24	0	0x80000000	0	0.0	Inbox (12) - unsub.fscs@gmail.com - Gmail
26	20	2022-04-19 13:58:40	25	0	0x80000000	0	0.0	Inbox (12) - unsub.fscs@gmail.com - Gmail
27	21	2022-04-19 13:58:40	26	0	0x80000000	0	0.0	Gmail: Free, Private & Secure Email Google
28	22	2022-04-19 13:58:40	27	0	0x80000000	0	0.0	Gmail: Free, Private & Secure Email Google
29	23	2022-04-19 13:58:40	28	0	0x80000000	0	0.0	Gmail: Free, Private & Secure Email Google
30	24	2022-04-19 13:58:40	29	0	0xA0000000	0	0.0	Gmail: Free, Private & Secure Email Google
31	25	2022-04-19 13:58:43	0	0	0x10000000	0	0.0	Gmail
32	26	2022-04-19 13:58:43	31	0	0x80000000	0	0.0	Gmail
33	27	2022-04-19 13:58:43	32	0	0xA0000000	0	0.0	Gmail
34	27	2022-04-19 13:58:44	0	33	0x30000000	0	0.0	Gmail
35	28	2022-04-19 13:58:44	0	34	0x30000000	0	0.0	Gmail
36	29	2022-04-19 13:59:26	0	35	0x30000000	0	3.0	Gmail
37	30	2022-04-19 13:59:29	36	0	0x10000000	0	0.0	Inbox (12) - unsub.fscs@gmail.com - Gmail

Figure 6a. Visit timeline in UTC, visits 1–37, with transition hex, core type and duration (Q04; E04). [file: screenshots/fig05a_timeline_1-37.png]

sqlite3 -readonly working/History_working (visits 38-74)

```
sqlite> .read sql/q04_timeline.sql
```

vid	uid	visit_utc	from_v	opener	trans_hex	core	dur_s	title
38	31	2022-04-19 13:59:29	37	0	0x80000000	0	0.0	Inbox (12) - unsub.fscs@gmail.com - Gmail
39	32	2022-04-19 13:59:29	38	0	0x80000000	0	0.0	Inbox (12) - unsub.fscs@gmail.com - Gmail
40	19	2022-04-19 13:59:29	39	0	0x80000000	0	0.0	Inbox (12) - unsub.fscs@gmail.com - Gmail
41	20	2022-04-19 13:59:29	40	0	0x80000000	0	0.0	Inbox (12) - unsub.fscs@gmail.com - Gmail
42	33	2022-04-19 13:59:31	41	41	0x60000000	0	0.0	Inbox (12) - unsub.fscs@gmail.com - Gmail
43	34	2022-04-19 14:04:12	0	42	0x10000000	0	0.0	cheaper than Rx supplements - order - unsub.
44	35	2022-04-19 14:04:35	43	43	0x40000000	0	0.0	cheaper than Rx supplements - order - unsub.
45	36	2022-04-19 14:04:46	44	44	0x40000000	0	0.0	cheaper than Rx supplements - order - unsub.
46	34	2022-04-19 14:04:52	45	45	0x60000000	0	0.0	cheaper than Rx supplements - order - unsub.
47	33	2022-04-19 14:05:59	0	46	0x30000000	0	0.0	Inbox (12) - unsub.fscs@gmail.com - Gmail
48	37	2022-04-19 14:56:47	0	47	0x10000000	0	0.0	cheaper than Rx supplements -- txid: 517b215
49	38	2022-04-19 14:56:52	0	48	0x10000000	0	0.0	Imgur: The magic of the Internet
50	39	2022-04-19 14:56:52	49	0	0xA0000000	0	18.9	Imgur: The magic of the Internet
51	40	2022-04-19 14:58:59	0	0	0x32000005	5	0.0	blockchain explorer - Google Search
52	40	2022-04-19 14:59:00	0	51	0x30000000	0	1.0	blockchain explorer - Google Search
53	41	2022-04-19 14:59:01	52	0	0x30000000	0	0.0	Blockchain Explorer - Search the Blockchain
54	41	2022-04-19 14:59:02	0	53	0x30000000	0	18.7	Blockchain Explorer - Search the Blockchain
55	42	2022-04-19 14:59:06	54	0	0x10000000	0	0.0	Transaction: 517b2156914944339a96137ad897840
56	43	2022-04-19 14:59:06	55	0	0xA0000000	0	0.0	Transaction: 517b2156914944339a96137ad897840
57	43	2022-04-19 14:59:06	54	56	0x30000000	0	0.0	Transaction: 517b2156914944339a96137ad897840
58	44	2022-04-19 14:59:21	54	57	0x30000000	0	0.0	Address: 38RcsURWCDmCbYocmsZCnGz1FtD7x477mt
59	45	2022-04-19 14:59:38	0	0	0x32000005	5	0.0	kraken - Google Search
60	45	2022-04-19 14:59:39	0	59	0x30000000	0	7.7	kraken - Google Search
61	46	2022-04-19 14:59:47	60	0	0x30000000	0	0.0	Sign in to Kraken - Kraken Buy, Sell and M
62	46	2022-04-19 14:59:47	0	61	0x30000000	0	0.0	Sign in to Kraken - Kraken Buy, Sell and M
63	47	2022-04-19 15:00:21	0	62	0x30000000	0	104.4	Approve new device - Kraken Buy, Sell and
64	48	2022-04-19 15:01:11	63	0	0x30000000	0	0.0	
65	49	2022-04-19 15:01:12	63	64	0x30000000	0	0.0	Kraken Buy, Sell and Margin Trade Bitcoin
66	50	2022-04-19 15:01:21	63	65	0x30000000	0	0.0	41,543.80 USD - Kraken - Funding - Overview
67	51	2022-04-19 15:01:31	0	66	0x10000000	0	0.0	Transaction: 517b2156914944339a96137ad897840
68	52	2022-04-19 15:01:33	67	0	0x40000000	0	0.0	Transaction: 517b2156914944339a96137ad897840
69	53	2022-04-19 15:01:33	68	0	0x80000000	0	0.0	Transaction: 517b2156914944339a96137ad897840
70	54	2022-04-19 15:01:33	69	0	0xA0000000	0	0.0	Transaction: 517b2156914944339a96137ad897840
71	55	2022-04-19 15:02:05	63	66	0x30000000	0	0.0	41,563.70 USD - Kraken - History - Ledger
72	56	2022-04-19 15:02:32	48	48	0x40000000	0	0.0	cheaper than Rx supplements - order - unsub.
73	57	2022-04-19 15:02:44	72	72	0x40000000	0	0.0	cheaper than Rx supplements - order - unsub.
74	37	2022-04-19 15:02:56	73	73	0x60000000	0	0.0	cheaper than Rx supplements -- txid: 517b215

Figure 6b. Visit timeline in UTC, visits 38-74 (Q04; E04). [file: screenshots/fig05b_timeline_38-74.png]

sqlite3 -readonly working/History_working (visits 1-37)

```
sqlite> .read sql/q05_transitions.sql
```

vid	from_v	trans_hex	core	qualifiers	url
1	0	0x30000005	GENERATED	CHAIN_START CHAIN_END	https://www.google.com/search?q=craigslist&rlz=1C10NGR_enU
2	0	0x30000000	LINK	CHAIN_START CHAIN_END	https://www.google.com/search?q=craigslist&rlz=1C10NGR_enU
3	2	0x10000000	LINK	CHAIN_START	https://www.craigslist.org/
4	3	0x80000000	LINK	SERVER_REDIRECT	https://geo.craigslist.org/
5	4	0xA0000000	LINK	CHAIN_END SERVER_REDIRECT	https://baltimore.craigslist.org/
6	5	0x10000000	LINK	CHAIN_START	https://accounts.craigslist.org/login/home
7	6	0xA0000000	LINK	CHAIN_END SERVER_REDIRECT	https://accounts.craigslist.org/login?rp=%2Flogin%2Fhome&r
8	7	0x30000007	FORM_SUBMIT	CHAIN_START CHAIN_END	https://accounts.craigslist.org/login/home
9	8	0x10000000	LINK	CHAIN_START	https://www.craigslist.org/
10	9	0xA0000000	LINK	CHAIN_END SERVER_REDIRECT	https://baltimore.craigslist.org/
11	10	0x10000000	LINK	CHAIN_START	https://post.craigslist.org/c/bal
12	11	0x80000000	LINK	SERVER_REDIRECT	https://post.craigslist.org/k/sHb0U0i_7BG68qUDIbeTbQ/bQTi5
13	12	0xA0000000	LINK	CHAIN_END SERVER_REDIRECT	https://post.craigslist.org/k/sHb0U0i_7BG68qUDIbeTbQ/bQTi5
14	13	0x30000007	FORM_SUBMIT	CHAIN_START CHAIN_END	https://post.craigslist.org/k/sHb0U0i_7BG68qUDIbeTbQ/bQTi5
15	14	0x30000007	FORM_SUBMIT	CHAIN_START CHAIN_END	https://post.craigslist.org/k/sHb0U0i_7BG68qUDIbeTbQ/bQTi5
16	15	0x30000007	FORM_SUBMIT	CHAIN_START CHAIN_END	https://post.craigslist.org/k/sHb0U0i_7BG68qUDIbeTbQ/bQTi5
17	16	0x30000007	FORM_SUBMIT	CHAIN_START CHAIN_END	https://post.craigslist.org/k/sHb0U0i_7BG68qUDIbeTbQ/bQTi5
18	17	0x30000007	FORM_SUBMIT	CHAIN_START CHAIN_END	https://post.craigslist.org/k/sHb0U0i_7BG68qUDIbeTbQ/bQTi5
19	18	0x30000007	FORM_SUBMIT	CHAIN_START CHAIN_END	https://post.craigslist.org/k/sHb0U0i_7BG68qUDIbeTbQ/bQTi5
20	19	0x30000000	LINK	CHAIN_START CHAIN_END	https://post.craigslist.org/manage/7473121658
21	0	0x30000000	LINK	CHAIN_START CHAIN_END	https://baltimore.craigslist.org/hab/d/baltimore-cheaper-t
22	0	0x32000005	GENERATED	FROM_ADDRESS_BAR CHAIN_START CHAIN_END	https://www.google.com/search?q=gmail&rlz=1C10NGR_enUS989U
23	0	0x30000000	LINK	CHAIN_START CHAIN_END	https://www.google.com/search?q=gmail&rlz=1C10NGR_enUS989U
24	23	0x10000000	LINK	CHAIN_START	https://www.google.com/gmail/
25	24	0x80000000	LINK	SERVER_REDIRECT	https://mail.google.com/mail/
26	25	0x80000000	LINK	SERVER_REDIRECT	https://mail.google.com/mail/u/0/
27	26	0x80000000	LINK	SERVER_REDIRECT	https://accounts.google.com/ServiceLogin?service=mail&pass
28	27	0x80000000	LINK	SERVER_REDIRECT	https://mail.google.com/intl/en/mail/help/about.html
29	28	0x80000000	LINK	SERVER_REDIRECT	https://www.google.com/intl/en/mail/help/about.html
30	29	0xA0000000	LINK	CHAIN_END SERVER_REDIRECT	https://www.google.com/gmail/about/
31	0	0x10000000	LINK	CHAIN_START	https://accounts.google.com/AccountChooser/signinchooser?s
32	31	0x80000000	LINK	SERVER_REDIRECT	https://accounts.google.com/AccountChooser?service=mail&co
33	32	0xA0000000	LINK	CHAIN_END SERVER_REDIRECT	https://accounts.google.com/ServiceLogin?continue=https%3A
34	0	0x30000000	LINK	CHAIN_START CHAIN_END	https://accounts.google.com/ServiceLogin?continue=https%3A
35	0	0x30000000	LINK	CHAIN_START CHAIN_END	https://accounts.google.com/signin/v2/identifier?continue=
36	0	0x30000000	LINK	CHAIN_START CHAIN_END	https://accounts.google.com/signin/v2/challenge/pwd?contin
37	36	0x10000000	LINK	CHAIN_START	https://accounts.google.com/CheckCookie?continue=https%3A%

Figure 7a. Decoded transition core types and qualifiers, visits 1-37 (Q05; E05). [file: screenshots/fig06a_transitions_1-37.png]

sqlite3 -readonly working/History_working (visits 38-74)

```
sqlite> .read sql/q05_transitions.sql
```

vid	from_v	trans_hex	core	qualifiers	url
38	37	0x80000000	LINK	SERVER_REDIRECT	https://mail.google.com/accounts/SetOSID?authuser=0&contin
39	38	0x80000000	LINK	SERVER_REDIRECT	https://accounts.youtube.com/accounts/SetSID?ssdc=1&sdt=A
40	39	0x80000000	LINK	SERVER_REDIRECT	https://mail.google.com/mail/
41	40	0x80000000	LINK	SERVER_REDIRECT	https://mail.google.com/mail/u/0/
42	41	0x60000000	LINK	CHAIN_END CLIENT_REDIRECT	https://mail.google.com/mail/u/0/#inbox
43	0	0x10000000	LINK	CHAIN_START	https://mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqXBGCn
44	43	0x40000000	LINK	CLIENT_REDIRECT	https://mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqXBGCn
45	44	0x40000000	LINK	CLIENT_REDIRECT	https://mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqXBGCn
46	45	0x60000000	LINK	CHAIN_END CLIENT_REDIRECT	https://mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqXBGCn
47	0	0x30000000	LINK	CHAIN_START CHAIN_END	https://mail.google.com/mail/u/0/#inbox
48	0	0x10000000	LINK	CHAIN_START	https://mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqvtrjh
49	0	0x10000000	LINK	CHAIN_START	https://www.google.com/url?q=https://imgur.com/dTgrkP7&sou
50	49	0xA0000000	LINK	CHAIN_END SERVER_REDIRECT	https://imgur.com/dTgrkP7
51	0	0x32000005	GENERATED	FROM_ADDRESS_BAR CHAIN_START CHAIN_END	https://www.google.com/search?q=blockchain+explorer&rlz=1C
52	0	0x30000000	LINK	CHAIN_START CHAIN_END	https://www.google.com/search?q=blockchain+explorer&rlz=1C
53	52	0x30000000	LINK	CHAIN_START CHAIN_END	https://www.blockchain.com/explorer
54	0	0x30000000	LINK	CHAIN_START CHAIN_END	https://www.blockchain.com/explorer
55	54	0x10000000	LINK	CHAIN_START	https://www.blockchain.com/search?search=517b2156914944339
56	55	0xA0000000	LINK	CHAIN_END SERVER_REDIRECT	https://www.blockchain.com/btc/tx/517b2156914944339a96137a
57	54	0x30000000	LINK	CHAIN_START CHAIN_END	https://www.blockchain.com/btc/tx/517b2156914944339a96137a
58	54	0x30000000	LINK	CHAIN_START CHAIN_END	https://www.blockchain.com/btc/address/38RcsURWCDmCbYocmsZ
59	0	0x32000005	GENERATED	FROM_ADDRESS_BAR CHAIN_START CHAIN_END	https://www.google.com/search?q=kraken&rlz=1C10NGR_enUS989
60	0	0x30000000	LINK	CHAIN_START CHAIN_END	https://www.google.com/search?q=kraken&rlz=1C10NGR_enUS989
61	60	0x30000000	LINK	CHAIN_START CHAIN_END	https://www.kraken.com/en-us/sign-in
62	0	0x30000000	LINK	CHAIN_START CHAIN_END	https://www.kraken.com/en-us/sign-in
63	0	0x30000000	LINK	CHAIN_START CHAIN_END	https://www.kraken.com/en-us/device-approval
64	63	0x30000000	LINK	CHAIN_START CHAIN_END	https://www.kraken.com/u/trade?i=true
65	63	0x30000000	LINK	CHAIN_START CHAIN_END	https://www.kraken.com/u/instant
66	63	0x30000000	LINK	CHAIN_START CHAIN_END	https://www.kraken.com/u/funding
67	0	0x10000000	LINK	CHAIN_START	https://www.kraken.com/redirect?url=https%3A%2F%2Fmempool.
68	67	0x40000000	LINK	CLIENT_REDIRECT	http://www.kraken.com/redirect?t=-_F22IhwssFD
69	68	0x80000000	LINK	SERVER_REDIRECT	https://www.kraken.com/redirect?t=-_F22IhwssFD
70	69	0xA0000000	LINK	CHAIN_END SERVER_REDIRECT	https://mempool.space/tx/517b2156914944339a96137ad8978408e
71	63	0x30000000	LINK	CHAIN_START CHAIN_END	https://www.kraken.com/u/history/ledger
72	48	0x40000000	LINK	CLIENT_REDIRECT	https://mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqvtrjh
73	72	0x40000000	LINK	CLIENT_REDIRECT	https://mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqvtrjh
74	73	0x60000000	LINK	CHAIN_END CLIENT_REDIRECT	https://mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqvtrjh

Figure 7b. Decoded transition core types and qualifiers, visits 38–74 (Q05; E05). [file: screenshots/fig06b_transitions_38-74.png]

sqlite3 -readonly working/History_working

```
sqlite> .read sql/q06_posting.sql
```

vid	uid	visit_utc	from_v	core	dur_s	title	url
3	2	2022-04-19 13:54:19	2	0	0.0	craigslist: baltimore jobs, apartments,	www.craigslist.org/
4	3	2022-04-19 13:54:19	3	0	0.0	craigslist: baltimore jobs, apartments,	geo.craigslist.org/
5	4	2022-04-19 13:54:19	4	0	2.5	craigslist: baltimore jobs, apartments,	baltimore.craigslist.org/
6	5	2022-04-19 13:54:22	5	0	0.0	craigslist account	accounts.craigslist.org/login/home
7	6	2022-04-19 13:54:22	6	0	17.5	craigslist - account log in	accounts.craigslist.org/login?rp=%2Flogin%2Fhome&rt=L
8	5	2022-04-19 13:54:39	7	7	23.4	craigslist account	accounts.craigslist.org/login/home
9	2	2022-04-19 13:55:03	8	0	0.0	craigslist: baltimore jobs, apartments,	www.craigslist.org/
10	4	2022-04-19 13:55:03	9	0	2.7	craigslist: baltimore jobs, apartments,	baltimore.craigslist.org/
11	7	2022-04-19 13:55:05	10	0	0.0	baltimore choose type	post.craigslist.org/c/bal
12	8	2022-04-19 13:55:05	11	0	0.0	baltimore posting confirmation	post.craigslist.org/k/sHb0U0i_7BG68qUDIbeTbQ/bQTi5
13	9	2022-04-19 13:55:05	12	0	4.3	baltimore choose type	post.craigslist.org/k/sHb0U0i_7BG68qUDIbeTbQ/bQTi5?s=type
14	10	2022-04-19 13:55:10	13	7	5.6	baltimore choose category	post.craigslist.org/k/sHb0U0i_7BG68qUDIbeTbQ/bQTi5?s=cat
15	11	2022-04-19 13:55:15	14	7	177.7	baltimore create posting	post.craigslist.org/k/sHb0U0i_7BG68qUDIbeTbQ/bQTi5?s=edit
16	12	2022-04-19 13:58:13	15	7	4.4	baltimore add map	post.craigslist.org/k/sHb0U0i_7BG68qUDIbeTbQ/bQTi5?s=geoverify
17	13	2022-04-19 13:58:17	16	7	1.6	baltimore create posting	post.craigslist.org/k/sHb0U0i_7BG68qUDIbeTbQ/bQTi5?s=editimage
18	14	2022-04-19 13:58:19	17	7	3.2	baltimore create posting	post.craigslist.org/k/sHb0U0i_7BG68qUDIbeTbQ/bQTi5?s=preview
19	8	2022-04-19 13:58:22	18	7	5.0	baltimore posting confirmation	post.craigslist.org/k/sHb0U0i_7BG68qUDIbeTbQ/bQTi5
20	15	2022-04-19 13:58:27	19	0	5.6	baltimore manage posting	post.craigslist.org/manage/7473121658
21	16	2022-04-19 13:58:31	0	0	0.0	cheaper than Rx supplements - health and	baltimore.craigslist.org/hab/d/baltimore-cheaper-than-rx-suppl

Figure 8. Craigslist sign-in and posting workflow, visits 3–21 (Q06; E07–E09). [file: screenshots/fig07_posting_workflow.png]

sqlite3 -readonly working/History_working

```
sqlite> .read sql/q07_gmail.sql
```

vid	uid	visit_utc	from_v	opener	trans_hex	title	view	thread_id_prefix
42	33	2022-04-19 13:59:31	41	41	0x60000000	Inbox (12) - unsub.fscs@gmail.com - Gmail	inbox	-
43	34	2022-04-19 14:04:12	0	42	0x10000000	cheaper than Rx supplements - order - unsub.fscs@gma	thread-view	WhctKKXXDrfCqXBG
44	35	2022-04-19 14:04:35	43	43	0x40000000	cheaper than Rx supplements - order - unsub.fscs@gma	compose-state	WhctKKXXDrfCqXBG
45	36	2022-04-19 14:04:46	44	44	0x40000000	cheaper than Rx supplements - order - unsub.fscs@gma	compose-state	WhctKKXXDrfCqXBG
46	34	2022-04-19 14:04:52	45	45	0x60000000	cheaper than Rx supplements - order - unsub.fscs@gma	thread-view	WhctKKXXDrfCqXBG
47	33	2022-04-19 14:05:59	0	46	0x30000000	Inbox (12) - unsub.fscs@gmail.com - Gmail	inbox	-
48	37	2022-04-19 14:56:47	0	47	0x10000000	cheaper than Rx supplements -- txid: 517b21569149443	thread-view	WhctKKXXDrfCqvtr
72	56	2022-04-19 15:02:32	48	48	0x40000000	cheaper than Rx supplements - order - unsub.fscs@gma	compose-state	WhctKKXXDrfCqvtr
73	57	2022-04-19 15:02:44	72	72	0x40000000	cheaper than Rx supplements - order - unsub.fscs@gma	compose-state	WhctKKXXDrfCqvtr
74	37	2022-04-19 15:02:56	73	73	0x60000000	cheaper than Rx supplements -- txid: 517b21569149443	thread-view	WhctKKXXDrfCqvtr

Figure 9. Gmail inbox, thread and compose views with thread-ID prefixes (Q07; E10–E12). [file: screenshots/fig08_gmail_threads.png]

```

sqlite3 -readonly working/History_working
sqlite> .read sql/q08_payment.sql
vid  uid  visit_utc  from_v  opener  trans_hex  title  url
---  ---  ---
49  38  2022-04-19 14:56:52  0  48  0x100000000  Imgur: The magic of the Internet  www.google.com/url?q=https://imgur.com/dTgrkP7&source=gmail&
50  39  2022-04-19 14:56:52  49  0  0xA00000000  Imgur: The magic of the Internet  imgur.com/dTgrkP7
53  41  2022-04-19 14:59:01  52  0  0x300000000  Blockchain Explorer - Search the Blockch  www.blockchain.com/explorer
54  41  2022-04-19 14:59:02  0  53  0x300000000  Blockchain Explorer - Search the Blockch  www.blockchain.com/explorer
55  42  2022-04-19 14:59:06  54  0  0x100000000  Transaction: 517b2156914944339a96137ad89  www.blockchain.com/search?search=517b2156914944339a96137ad89
56  43  2022-04-19 14:59:06  55  0  0xA00000000  Transaction: 517b2156914944339a96137ad89  www.blockchain.com/btc/tx/517b2156914944339a96137ad8978408ea
57  43  2022-04-19 14:59:06  54  56  0x300000000  Transaction: 517b2156914944339a96137ad89  www.blockchain.com/btc/tx/517b2156914944339a96137ad8978408ea
58  44  2022-04-19 14:59:21  54  57  0x300000000  Address: 38RcsURWCDmCbYocmsZCnGz1FtD7x47  www.blockchain.com/btc/address/38RcsURWCDmCbYocmsZCnGz1FtD7x
61  46  2022-04-19 14:59:47  60  0  0x300000000  Sign in to Kraken - Kraken | Buy, Sell a  www.kraken.com/en-us/sign-in
62  46  2022-04-19 14:59:47  0  61  0x300000000  Sign in to Kraken - Kraken | Buy, Sell a  www.kraken.com/en-us/sign-in
63  47  2022-04-19 15:00:21  0  62  0x300000000  Approve new device - Kraken | Buy, Sell  www.kraken.com/en-us/device-approval
64  48  2022-04-19 15:01:11  63  0  0x300000000  www.kraken.com/u/trade?i=true
65  49  2022-04-19 15:01:12  63  64  0x300000000  Kraken | Buy, Sell and Margin Trade Bitc  www.kraken.com/u/instant
66  50  2022-04-19 15:01:21  63  65  0x300000000  41,543.80 USD - Kraken - Funding - Overv  www.kraken.com/u/funding
67  51  2022-04-19 15:01:31  0  66  0x100000000  Transaction: 517b2156914944339a96137ad89  www.kraken.com/redirect?url=https%3A%2F%2Fmempool.space%2Ftx
68  52  2022-04-19 15:01:33  67  0  0x400000000  Transaction: 517b2156914944339a96137ad89  www.kraken.com/redirect?t=-_F22IhwssFD
69  53  2022-04-19 15:01:33  68  0  0x800000000  Transaction: 517b2156914944339a96137ad89  www.kraken.com/redirect?t=-_F22IhwssFD
70  54  2022-04-19 15:01:33  69  0  0xA00000000  Transaction: 517b2156914944339a96137ad89  mempool.space/tx/517b2156914944339a96137ad8978408ea52b2fc144
71  55  2022-04-19 15:02:05  63  66  0x300000000  41,563.70 USD - Kraken - History - Ledger  www.kraken.com/u/history/ledger

```

Figure 10. Imgur, blockchain.com, Kraken and mempool.space visits (Q08; E13, E15–E17). [file: screenshots/fig09_imgur_blockchain_kraken.png]

```

sqlite3 -readonly working/History_working
sqlite> .read sql/q09_txid.sql
vid  uid  visit_utc  in_url  in_title  host_path
---  ---  ---
48  37  2022-04-19 14:56:47  -  Y  mail.google.com/mail/u/0/#inbox/Wh
55  42  2022-04-19 14:59:06  Y  Y  www.blockchain.com/search?search=5
56  43  2022-04-19 14:59:06  Y  Y  www.blockchain.com/btc/tx/517b2156
57  43  2022-04-19 14:59:06  Y  Y  www.blockchain.com/btc/tx/517b2156
67  51  2022-04-19 15:01:31  Y  Y  www.kraken.com/redirect?url=https%
68  52  2022-04-19 15:01:33  -  Y  ww.kraken.com/redirect?t=-_F22Ihws
69  53  2022-04-19 15:01:33  -  Y  www.kraken.com/redirect?t=-_F22Ihw
70  54  2022-04-19 15:01:33  Y  Y  mempool.space/tx/517b2156914944339
74  37  2022-04-19 15:02:56  -  Y  mail.google.com/mail/u/0/#inbox/Wh

```

Figure 11. All visits whose URL or title contains the transaction ID (Q09; E12, E15, E17). [file: screenshots/fig10_txid_correlation.png]

```

sqlite3 -readonly working/History_working
sqlite> .mode line
sqlite> .read sql/q10_downloads.sql
      id = 1
target_path = C:\Users\FSCS_User\Desktop\proof_of_payment.png
start_utc = 2022-04-19 14:56:58.494
end_utc = 2022-04-19 14:57:10.261
  rcvd = 33844
  total = 33844
  state = 1
  danger = 0
  opened = 0
mime_type = image/png
referrer = https://imgur.com/
tab_url = https://imgur.com/dTgrkP7
last_modified = Tue, 19 Apr 2022 14:52:06 GMT
chain_index = 0
chain_url = https://i.imgur.com/dTgrkP7.png

```

Figure 12. Download record joined to downloads_url_chains, sub-second UTC (Q10; E14). [file: screenshots/fig11_download_record.png]

```

sqlite3 -readonly working/History_working
sqlite> .read sql/q11_searches.sql
url_id  term                visit_count  typed_count  last_visit_utc
-----
1       craigslist          2            0            2022-04-19 13:54:19
17      gmail               2            0            2022-04-19 13:58:39
40      blockchain explorer 2            0            2022-04-19 14:59:00
45      kraken              2            0            2022-04-19 14:59:39

```

Figure 13. keyword_search_terms joined to urls (Q11; E06). [file: screenshots/fig12_search_terms.png]

```

sqlite3 -readonly working/History_working
sqlite> .read sql/q12_chain.sql
chain_for  depth  vid  trans_hex  core  url
-----
20         18     2    0x30000000  0     www.google.com/search?q=craigslist&rlz=1C10NGR_enUS989US989&
20         17     3    0x10000000  0     www.craigslist.org/
20         16     4    0x80000000  0     geo.craigslist.org/
20         15     5    0xA0000000  0     baltimore.craigslist.org/
20         14     6    0x10000000  0     accounts.craigslist.org/login/home
20         13     7    0xA0000000  0     accounts.craigslist.org/login?rp=%2Flogin%2Fhome&rt=L
20         12     8    0x30000007  7     accounts.craigslist.org/login/home
20         11     9    0x10000000  0     www.craigslist.org/
20         10    10    0xA0000000  0     baltimore.craigslist.org/
20          9    11    0x10000000  0     post.craigslist.org/c/bal
20          8    12    0x80000000  0     post.craigslist.org/k/sHb0U0i_7BG68qUDIbeTbQ/bQTi5
20          7    13    0xA0000000  0     post.craigslist.org/k/sHb0U0i_7BG68qUDIbeTbQ/bQTi5?s=type
20          6    14    0x30000007  7     post.craigslist.org/k/sHb0U0i_7BG68qUDIbeTbQ/bQTi5?s=cat
20          5    15    0x30000007  7     post.craigslist.org/k/sHb0U0i_7BG68qUDIbeTbQ/bQTi5?s=edit
20          4    16    0x30000007  7     post.craigslist.org/k/sHb0U0i_7BG68qUDIbeTbQ/bQTi5?s=geoveri
20          3    17    0x30000007  7     post.craigslist.org/k/sHb0U0i_7BG68qUDIbeTbQ/bQTi5?s=editima
20          2    18    0x30000007  7     post.craigslist.org/k/sHb0U0i_7BG68qUDIbeTbQ/bQTi5?s=preview
20          1    19    0x30000007  7     post.craigslist.org/k/sHb0U0i_7BG68qUDIbeTbQ/bQTi5
20          0    20    0x30000000  0     post.craigslist.org/manage/7473121658
70          3    67    0x10000000  0     www.kraken.com/redirect?url=https%3A%2F%2Fmempool.space%2Ftx
70          2    68    0x40000000  0     www.kraken.com/redirect?t=-_F22IhwssFD
70          1    69    0x80000000  0     www.kraken.com/redirect?t=-_F22IhwssFD
70          0    70    0xA0000000  0     mempool.space/tx/517b2156914944339a96137ad8978408ea52b2fc144

```

Figure 14. Recursive from_visit walk for visit 20 (posting) and visit 70 (mempool) (Q12; E05, E08, E17). [file: screenshots/fig13_from_visit_chains.png]

```

sqlite3 -readonly working/History_working
sqlite> .read sql/q13_typed.sql
total_typed_count  typed_visits  first_utc  last_utc
-----
0                  0          2022-04-19 13:54:18  2022-04-19 15:02:56

```

Figure 15. Typed-navigation check and first/last visit times (Q13; E04, E06). [file: screenshots/fig14_typed_and_window.png]

```

sqlite3 -readonly working/History_working
sqlite> .read sql/q14_integrity.sql
visit_ids  url_ids      urls_seq
-----
1-74 n=74  1-57 n=57  57
freelist_count
-----
0
page_count
-----
41

```

Figure 16. Record-ID contiguity, sqlite_sequence and free-page count (Q14; E18). [file: screenshots/fig16_integrity_checks.png]

```

sqlite3 -readonly working/History_working
sqlite> .read sql/q15_transition_counts.sql
trans_hex  core  n  visit_ids
-----
0x30000000 0    21  2,20,21,23,34,35,36,47,52,53,54,57,58,60,61,62,63,64,65,66,71
0x80000000 0    13  4,12,25,26,27,28,29,32,38,39,40,41,69
0x10000000 0    12  3,6,9,11,24,31,37,43,48,49,55,67
0xA0000000 0    9   5,7,10,13,30,33,50,56,70
0x30000007 7    7   8,14,15,16,17,18,19
0x40000000 0    5   44,45,68,72,73
0x60000000 0    3   42,46,74
0x32000005 5    3   22,51,59
0x30000005 5    1    1

```

Figure 17. Frequency of each transition value (Q15; E05). [file: screenshots/fig17_transition_counts.png]

12.4 SQL Query Listing

Q01–Q03 are the schema queries shown in Section 4 (SELECT name FROM sqlite_master ..., SELECT key, value FROM meta, PRAGMA table_info(...)).

Q04–Q15 follow. Each was executed as `sqlite3 -readonly -header -column working/History_working < sql/<file>.sql > reports/<file>.txt`.

sql/q04_timeline.sql

```
-- Q04: Full visit timeline (UTC) with decoded transition
SELECT v.id AS vid, u.id AS uid,
       datetime(v.visit_time/1000000 + strftime('%s','1601-01-01'),'unixepoch') AS visit_utc,
       v.from_visit AS from_v, v.opener_visit AS opener,
       printf('0x%08X', v.transition & 0xFFFFFFFF) AS trans_hex,
       v.transition & 0xFF AS core,
       round(v.visit_duration/1000000.0,1) AS dur_s,
       substr(u.title,1,44) AS title
FROM visits v JOIN urls u ON v.url = u.id
ORDER BY v.visit_time, v.id;
```

sql/q05_transitions.sql

```
-- Q05: Transition core type and qualifier decoding
SELECT v.id AS vid, v.from_visit AS from_v,
       printf('0x%08X', v.transition & 0xFFFFFFFF) AS trans_hex,
       CASE v.transition & 0xFF WHEN 0 THEN 'LINK' WHEN 1 THEN 'TYPED'
         WHEN 5 THEN 'GENERATED' WHEN 7 THEN 'FORM_SUBMIT' WHEN 8 THEN 'RELOAD'
         ELSE 'OTHER' END AS core,
       trim(
         CASE WHEN v.transition & 0x02000000 THEN 'FROM_ADDRESS_BAR ' ELSE '' END ||
         CASE WHEN v.transition & 0x10000000 THEN 'CHAIN_START ' ELSE '' END ||
         CASE WHEN v.transition & 0x20000000 THEN 'CHAIN_END ' ELSE '' END ||
         CASE WHEN v.transition & 0x40000000 THEN 'CLIENT_REDIRECT ' ELSE '' END ||
         CASE WHEN (v.transition & 0xFFFFFFFF) & 0x80000000 THEN 'SERVER_REDIRECT' ELSE '' END) AS qualifiers,
       substr(u.url,1,58) AS url
FROM visits v JOIN urls u ON v.url = u.id
ORDER BY v.visit_time, v.id;
```

sql/q06_posting.sql

```
-- Q06: Craigslist account and posting workflow (visits 1-21)
SELECT v.id AS vid, u.id AS uid,
       datetime(v.visit_time/1000000 + strftime('%s','1601-01-01'),'unixepoch') AS visit_utc,
       v.from_visit AS from_v, v.transition & 0xFF AS core,
       round(v.visit_duration/1000000.0,1) AS dur_s,
       substr(u.title,1,40) AS title,
       substr(u.url, instr(u.url,'://')+3, 62) AS url
FROM visits v JOIN urls u ON v.url = u.id
WHERE u.url LIKE '%craigslist.org%'
```

```
ORDER BY v.visit_time, v.id;
```

sql/q07_gmail.sql

```
-- Q07: Gmail communication artefacts (case-relevant thread visits)
SELECT v.id AS vid, u.id AS uid,
       datetime(v.visit_time/1000000 + strftime('%s','1601-01-01'),'unixepoch') AS visit_utc,
       v.from_visit AS from_v, v.opener_visit AS opener,
       printf('0x%08X', v.transition & 0xFFFFFFFF) AS trans_hex,
       substr(u.title,1,52) AS title,
       CASE WHEN u.url LIKE '%compose=%' THEN 'compose-state'
            WHEN u.url LIKE '%#inbox/%' THEN 'thread-view'
            WHEN u.url LIKE '%#inbox' THEN 'inbox' END AS view,
       CASE WHEN instr(u.url,'#inbox/')>0 THEN substr(u.url, instr(u.url,'#inbox/')+7, 16) ELSE '-' END AS thread_id_prefix
FROM visits v JOIN urls u ON v.url = u.id
WHERE u.url LIKE 'https://mail.google.com/mail/u/0/#inbox%'
ORDER BY v.visit_time, v.id;
```

sql/q08_payment.sql

```
-- Q08: Image-hosting, blockchain-explorer and exchange artefacts (visits 48-71)
SELECT v.id AS vid, u.id AS uid,
       datetime(v.visit_time/1000000 + strftime('%s','1601-01-01'),'unixepoch') AS visit_utc,
       v.from_visit AS from_v, v.opener_visit AS opener,
       printf('0x%08X', v.transition & 0xFFFFFFFF) AS trans_hex,
       substr(u.title,1,40) AS title,
       substr(u.url, instr(u.url,'://')+3, 60) AS url
FROM visits v JOIN urls u ON v.url = u.id
WHERE u.url LIKE '%imgur%' OR u.url LIKE '%blockchain.com%'
      OR u.url LIKE '%kraken.com%' OR u.url LIKE '%mempool.space%'
ORDER BY v.visit_time, v.id;
```

sql/q09_txid.sql

```
-- Q09: Every URL or title containing the transaction ID
SELECT v.id AS vid, u.id AS uid,
       datetime(v.visit_time/1000000 + strftime('%s','1601-01-01'),'unixepoch') AS visit_utc,
       CASE WHEN u.url LIKE '%517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5%' THEN 'Y' ELSE '-' END AS in_url,
       CASE WHEN u.title LIKE '%517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5%' THEN 'Y' ELSE '-' END AS in_title,
       substr(u.url,9,34) AS host_path
FROM visits v JOIN urls u ON v.url = u.id
WHERE u.url LIKE '%517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5%'
      OR u.title LIKE '%517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5%'
ORDER BY v.visit_time, v.id;
```

sql/q10_downloads.sql

```
-- Q10: Download record with URL chain (UTC)
SELECT d.id, d.target_path,
       strftime('%Y-%m-%d %H:%M:%f', d.start_time/1000000.0 - 11644473600,'unixepoch') AS start_utc,
       strftime('%Y-%m-%d %H:%M:%f', d.end_time/1000000.0 - 11644473600,'unixepoch') AS end_utc,
       d.received_bytes AS rcvd, d.total_bytes AS total, d.state, d.danger_type AS danger,
```

```

    d.opened, d.mime_type, d.referrer, d.tab_url, d.last_modified,
    c.chain_index, c.url AS chain_url
FROM downloads d LEFT JOIN downloads_url_chains c ON c.id = d.id
ORDER BY d.start_time;

```

sql/q11_searches.sql

```

-- Q11: Search terms recorded by the browser, joined to the search-results URL
SELECT k.url_id, k.term, u.visit_count, u.typed_count,
       datetime(u.last_visit_time/1000000 + strftime('%s','1601-01-01'),'unixepoch') AS last_visit_utc
FROM keyword_search_terms k JOIN urls u ON u.id = k.url_id
ORDER BY u.last_visit_time;

```

sql/q12_chain.sql

```

-- Q12: Walk from visit back from the manage-posting visit (20) and the mempool visit (70)
WITH RECURSIVE chain(start, vid, depth) AS (
    SELECT id, id, 0 FROM visits WHERE id IN (20, 70)
    UNION ALL
    SELECT c.start, v.from_visit, c.depth+1 FROM chain c JOIN visits v ON v.id = c.vid
    WHERE v.from_visit <> 0)
SELECT c.start AS chain_for, c.depth, v.id AS vid,
       printf('0x%08X', v.transition & 0xFFFFFFFF) AS trans_hex,
       v.transition & 0xFF AS core,
       substr(u.url, instr(u.url,'://')+3, 60) AS url
FROM chain c JOIN visits v ON v.id = c.vid JOIN urls u ON u.id = v.url
ORDER BY c.start, c.depth DESC;

```

sql/q13_typed.sql

```

-- Q13: Typed-navigation check and time window
SELECT SUM(typed_count) AS total_typed_count,
       (SELECT COUNT(*) FROM visits WHERE transition & 0xFF = 1) AS typed_visits,
       (SELECT datetime(MIN(visit_time)/1000000 + strftime('%s','1601-01-01'),'unixepoch') FROM visits) AS first_utc,
       (SELECT datetime(MAX(visit_time)/1000000 + strftime('%s','1601-01-01'),'unixepoch') FROM visits) AS last_utc
FROM urls;

```

sql/q14_integrity.sql

```

-- Q14: Record-ID contiguity and free-page check (deletion indicators)
SELECT (SELECT MIN(id)||'-'||MAX(id)||' n='||COUNT(*) FROM visits) AS visit_ids,
       (SELECT MIN(id)||'-'||MAX(id)||' n='||COUNT(*) FROM urls) AS url_ids,
       (SELECT seq FROM sqlite_sequence WHERE name='urls') AS urls_seq;
PRAGMA freelist_count;
PRAGMA page_count;

```

sql/q15_transition_counts.sql

```

-- Q15: Frequency of each transition value
SELECT printf('0x%08X', transition & 0xFFFFFFFF) AS trans_hex, transition & 0xFF AS core,
       COUNT(*) AS n, group_concat(id) AS visit_ids
FROM visits GROUP BY 1 ORDER BY n DESC;

```